3

Stochastic Processes

1. INTRODUCTION

A scalar (vector) **stochastic process** $\{x_t, t \in T\}$ is a family of random variables (vectors) indexed by the parameter set T.[1] The words *stochastic* and *random* are used synonymously here. The parameter t will refer to *time* in our applications. If the random variables (vectors) x_t are discrete, we say that the stochastic process has a *discrete state space*. If they are continuous, the process is said to have a *continuous state space*. The parameter set T may also be discrete ($T = \{1, 2,..., n\}$, $T = \{1, 2,...\}$) or continuous ($T = [0, 1]$, $T = \{t : t \geqslant 0\}$). If the parameter set is discrete, the stochastic process is a *discrete parameter process*; if it is continuous, we say that the stochastic process is a *continuous parameter process*. A diagram of the classification we have just described appears in Table 3.1.

[1] In Chapter 2, we made a notational distinction between random variables and their realizations. Lower-case roman letters were used for the former, and lower-case italic letters for the latter. This, we felt, was helpful to a reader not acquainted with the theory. The reader should have enough practice by now to make this distinction unnecessary. Whether we are talking about an rv or its sample value should be clear from the context. Henceforth, therefore, we shall use lower-case italic characters for random variables (and vectors) as well as their realizations. Furthermore, when there is no possible confusion, we shall drop the cumbersome subscripts on distribution and density functions. The argument will identify the rv under discussion.

TABLE 3.1

CLASSIFICATION OF STOCHASTIC PROCESSES

State space	Parameter set: Discrete	Parameter set: Continuous
Continuous	Random sequence	(Stochastic) process Random function
Discrete	Discrete parameter chain	Continuous parameter chain

Except for isolated examples, we shall be concerned here *only* with continuous state space, discrete parameter and continuous parameter stochastic processes. We shall refer to continuous state space, discrete parameter processes as **random sequences**. A continuous state space, continuous parameter process will often be called a **stochastic process** or simply a **process**, when no confusion is likely to arise. Such a process will sometimes be called a **random function**.

Note that a stochastic process is really a function of two variables, the *time parameter* t and the *probability parameter* ω. The notation $\{x_t, t \in T\}$ is shorthand for $\{x_t(\omega), t \in T, \omega \in \Omega\}$. For each t, $x_t(\cdot)$ is a random variable (vector). For each ω, $x.(\omega)$ is a **realization** of the process: a sample function, if the parameter set is continuous; and a sample sequence, if it is discrete. $x_t(\omega)$ is sometimes written $x(t, \omega)$, although we shall not use this notation here. The inclusion of the argument t in parentheses [as in $y(t)$] will be reserved for fixed (nonrandom) functions.

Example 3.1 (*Random Walk*). Suppose that, at time instants 0, 1, 2,..., we toss a coin. If heads (h) comes up, we take a step $+\Delta x$ forward, and if tails (t) comes up, we take a $-\Delta x$ step backward ($\Delta x \geqslant 0$). Assume that we execute the steps instantaneously. Let $\Omega = \{h, t\}$. Let $\Pr\{h\} = p$, $\Pr\{t\} = q = 1 - p$ ($p = q = \frac{1}{2}$ for a *fair* coin). For each time instant, define the random variable

$$w_n(\omega) = \begin{cases} +\Delta x, & \omega = h \\ -\Delta x, & \omega = t, \end{cases} \tag{3.1}$$

so that

$$\Pr\{w_n(\omega) = +\Delta x\} = p, \qquad \Pr\{w_n(\omega) = -\Delta x\} = q. \tag{3.2}$$

Let x_n denote our position at instant n, but before the execution of a step at that instant. Assuming we start at the origin, $x_0 = 0$, our position at n is given by

$$x_n = \sum_{i=0}^{n-1} w_i, \qquad n = 1, 2, \ldots . \tag{3.3}$$

It is clear that $\{x_n, n = 0, 1, \ldots\}$ is a **discrete parameter chain**. For each n, $x_n(\cdot)$ is a discrete rv given by the sum of rv's in (3.3). Each realization of the chain is a sequence of real numbers of the form $\pm k\,\Delta x$, $k = 0, 1, \ldots$. If we observe the process in continuous time, each realization is a staircase-type function. One realization of the chain is shown in Fig. 3.1. The stochastic process we have generated is called a (simple)

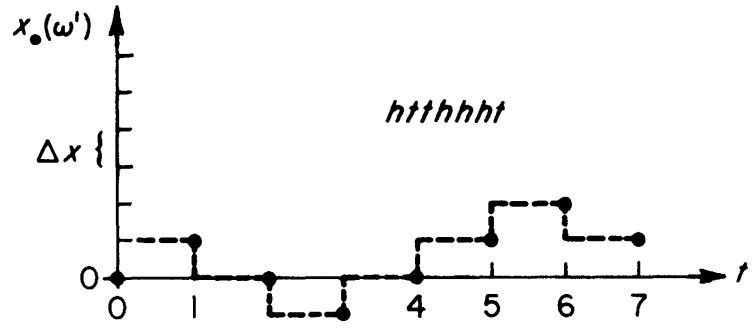


FIG. 3.1. Random walk.

random walk. Since we have defined the random walk as a function of the rv's w_i, we need not concern ourselves about the probability space associated with the walk. It is interesting to note that Ω', the probability space on which the random walk is defined, is the space of sequences of the form

$$\omega' = htthhht \cdots.$$

Now the statistics (probability law) of the w_n could be different. For example, the w_n could be specified to be continuous, independent, each identically distributed with density function $p(w)$. In that case, the process $\{x_n, n = 0, 1, \ldots\}$ generated by (3.3) would be a **random sequence**. For each n, $x_n(\cdot)$ would be a continuous rv. Realizations would be real sequences. Note that (3.3) is the solution of the *difference equation*

$$x_{n+1} = x_n + w_n; \qquad n = 0, 1, \ldots; \qquad x_0 = 0. \tag{3.4}$$

This is a *random* difference equation [see Section 9]. Its solution is a random sequence. ▲

Example 3.2. Consider the emission of particles by a radiating source. The emission can be considered to be continuous on a macroscopic scale. Let x_t be the number of particles emitted in the time interval $[0, t]$ (the *population* at time t). Then $\{x_t, t \geqslant 0\}$ is a **continuous parameter chain,** since x_t takes on integer values. ▲

Example 3.3. Consider a satellite in a periodic orbit around the earth. Not all forces acting on the satellite are known. We may, in fact, describe the satellite motion via a differential equation with a random forcing term: a *stochastic* differential equation. The altitude of the satellite, h_t, which is part of the solution of that equation, is a (continuous state space, continuous parameter) **stochastic process** $\{h_t, t \in [0, \pi]\}$, where π is the orbital period. The realizations of the process (altitude traces) on the kth and the $(k + 1)$st pass are shown in Fig. 3.2. ▲

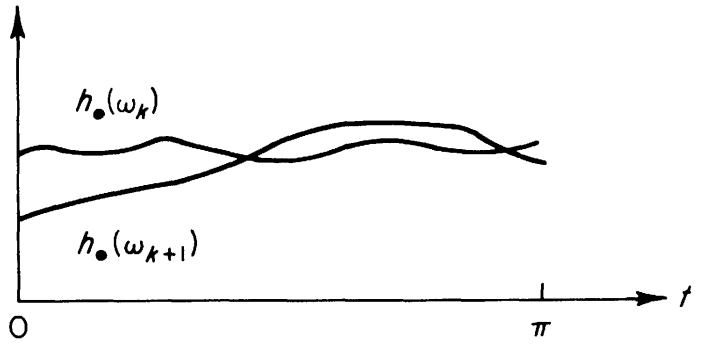


FIG. 3.2. Satellite altitude.

Our treatment of stochastic processes is limited to those concepts needed for the study of stochastic difference and differential equations. More complete formal studies, with numerous applications in various fields, can be found in Parzen [11], Papoulis [10], Cox and Miller [4], and Bartlett [2]. Consult Doob [5] for a complete mathematical treatment.

2. PROBABILITY LAW OF A STOCHASTIC PROCESS

Let $\{x_t, t \in T\}$ be a stochastic process. Its parameter set T may be discrete or continuous. For any finite set $\{t_1, \ldots, t_n\} \triangleq \{t_i\} \in T$, the joint distribution function of the random variables (vectors) $x_{t_1}, \ldots, x_{t_n}$ is called a **finite dimensional distribution** of the process. We state without proof that a stochastic process can be characterized (or statistically determined) by specifying the finite dimensional distribution

$$F(x_{t_1}, \ldots, x_{t_n}) \tag{3.5}$$

for *all* finite sets $\{t_i\} \in T$. By this we mean that, with the distribution functions (3.5) in hand, we can answer all probabilistic questions about the process. Equivalently, the stochastic process can be characterized by specifying the **joint density function**

$$p(x_{t_1}, \ldots, x_{t_n}) \tag{3.6}$$

or the **joint characteristic function**

$$\varphi_{x_{t_1}, \ldots, x_{t_n}}(u_1, \ldots, u_n) \tag{3.7}$$

for *all* finite sets $\{t_i\} \in T$. To specify (3.5), (3.6), or (3.7) for *all* finite sets $\{t_i\} \in T$ is to specify the **probability law** of the stochastic process.

Example 3.4. A sequence of independent rv's $\{w_0, w_1, \ldots\}$ constitutes a random sequence whose parameter set is $T = \{0, 1, \ldots\}$. Since the rv's are independent,

$$\varphi_{w_{m_1}, \ldots, w_{m_n}}(u_1, \ldots, u_n) = \varphi_{w_{m_1}}(u_1) \cdots \varphi_{w_{m_n}}(u_n)$$

for all $\{m_1, \ldots, m_n\} \in T$. So, to specify the probability law of this process, it suffices to specify

$$\{\varphi_{w_k}(u_k), k = 0, 1, \ldots\}.$$

If, moreover, the w_k are identically distributed, it suffices to specify one characteristic function $\varphi_w(u)$. ▲

Example 3.5. A stochastic process $\{x_t, t \geqslant 0\}$ may be defined by an equation such as

$$x_t = a + bt \tag{3.8}$$

or

$$x_t = a \cos 2\pi t + b \sin 2\pi t, \tag{3.9}$$

where a and b are random variables with specified distribution. The sample functions of the process defined by (3.8) are straight lines, whereas those of the process defined by (3.9) are sine waves. We say that these equations *generate* a stochastic process. Suppose a and b are jointly normally distributed. We can write, from (3.8),

$$\begin{bmatrix} x_{t_1} \\ \vdots \\ x_{t_n} \end{bmatrix} = \begin{bmatrix} 1, & t_1 \\ \vdots & \vdots \\ 1, & t_n \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix}. \tag{3.10}$$

A similar relationship can be obtained from (3.9), with an appropriately different coefficient matrix. Equation (3.10) represents a linear transformation of the Gaussian random vector $[a, b]^T$, and, by Theorem 2.11, $[x_{t_1}, \ldots, x_{t_n}]^T$ is a Gaussian random vector with mean and covariance matrix which can be written simply in terms of the mean and covariance matrix of $[a, b]^T$. Therefore, we can write down the probability law of the $\{x_t, t \geqslant 0\}$ process. ▲

Example 3.6. It was seen in Example 3.1 that the stochastic difference equation (3.4) generates a random sequence:

$$x_n = \sum_{i=0}^{n-1} w_i, \qquad n = 1, 2, \ldots . \tag{3.3}$$

Recall that the w_i are independent. Suppose at first that they are not identically distributed, but that the characteristic functions $\varphi_{w_i}(u_i)$, $i = 0, 1, \ldots$, are given. We can write

$$\begin{aligned} \sum_{i=1}^{n} u_i x_i &= u_n(x_n - x_{n-1}) + (u_{n-1} + u_n)(x_{n-1} - x_{n-2}) \\ &\quad + \cdots + (u_1 + \cdots + u_n)\, x_1 \\ &= u_n w_{n-1} + (u_{n-1} + u_n)\, w_{n-2} \\ &\quad + \cdots + (u_1 + \cdots + u_n)\, w_0 . \end{aligned} \tag{3.11}$$

Therefore,

$$\varphi_{x_1,\ldots,x_n}(u_1, \ldots, u_n) = \varphi_{w_{n-1}}(u_n)\, \varphi_{w_{n-2}}(u_{n-1} + u_n) \cdots \varphi_{w_0}(u_1 + \cdots + u_n), \tag{3.12}$$

and the probability law of the x_n random sequence is specified. If, moreover, the w_i are identically distributed, it suffices to specify the one characteristic function $\varphi_w(u)$ or, equivalently, one density function $p(w)$. ▲

Let $\{x_t, t \in T\}$ be a (discrete or continuous parameter scalar or vector) stochastic process. Although, in general, insufficient for the specification of the probability law of the process, the densities[2]

$$p(x_t), \qquad p(x_t, x_\tau), \qquad t, \tau \in T \tag{3.13}$$

will play a fundamental role in this book.[3] The first density function in

[2] As stated in Chapter 2, we assume that all the densities that we write down exist.

[3] They can, of course, be obtained from (3.6) as marginal densities. Similarly, the first density in (3.13) can be obtained from the second.

(3.13) is called the **first-order** density of the process. The second one is called the **second-order** density of the process. Note that, in general, $p(x_t)$ is a function of t, whereas $p(x_t, x_\tau)$ is a function of t and τ. To emphasize this, we may write

$$p(x_t) = p(x, t). \tag{3.14}$$

For each t, $p(\cdot, t)$ is the density $p_{x_t}(\cdot)$ of the random variable (vector) $x_t(\cdot)$. Similarly,

$$p(x_t, x_\tau) = p(x, y, t, \tau). \tag{3.15}$$

For each t and τ, $p(\cdot, \cdot, t, \tau)$ is the joint density $p_{x_t, x_\tau}(\cdot, \cdot)$ of the random variables (vectors) $x_t(\cdot)$ and $x_\tau(\cdot)$. The variables x and y in (3.14) and (3.15) are *dummy* variables. One may think of x as a realization of the rv x_t, and of y as a realization of x_τ. The *conditional* density

$$p(x_t \mid x_\tau) = p(x_t, x_\tau)/p(x_\tau) \tag{3.16}$$

has been defined in (2.100). $p(x_t \mid x_\tau)$ is also a function of t and τ. We sometimes write

$$p(x_t \mid x_\tau) = p(x, t; y, \tau). \tag{3.17}$$

At the risk of boring the reader, we write

$$p_{x_t \mid x_\tau}(x \mid y) = p(x_t \mid x_\tau)$$

to show the relationship with the subscript notation used in Chapter 2. Equation (3.16) can be written as

$$p(x, t; y, \tau) = p(x, y, t, \tau)/p(y, \tau). \tag{3.16}$$

The first- and second-order densities of a stochastic process can answer many important (probabilistic) questions about the process. For two important classes of stochastic processes (Gaussian processes, Section 6; and Markov processes, Section 7), they answer *all* the questions about the process. That is, they specify the probability law of the process.

We now define some parameters (statistics) associated with the first- and second-order densities of a stochastic process. Let $\{x_t, t \in T\}$ be a (discrete or continuous parameter) scalar process. The t function

$$m_x(t) \triangleq \mathscr{E}\{x_t\} \tag{3.18}$$

is called the **mean value function** of the process. The t, τ function

$$\gamma_{xx}(t, \tau) \triangleq \mathscr{E}\{x_t x_\tau\} \tag{3.19}$$

is called the **(auto) correlation function** of the process. The t, τ function

$$c_x(t, \tau) \triangleq \mathscr{E}\{[x_t - m_x(t)][x_\tau - m_x(\tau)]\} \tag{3.20}$$

is the **(auto) covariance function** (*covariance kernel*) of the process. We have that

$$c_x(t, \tau) = \gamma_x(t, \tau) - m_x(t)\, m_x(\tau). \tag{3.21}$$

In case $\{x_t, t \in T\}$ is an n-vector process, we have the **mean value vector** (function)

$$m_x(t) = \mathscr{E}\{x_t\}, \tag{3.18}$$

the **(auto-cross) correlation matrix**

$$\Gamma_x(t, \tau) = \mathscr{E}\{x_t x_\tau^{\mathrm{T}}\}, \tag{3.22}$$

and the **(auto-cross) covariance matrix**

$$C_x(t, \tau) = \mathscr{E}\{[x_t - m_x(t)][x_\tau - m_x(\tau)]^{\mathrm{T}}\} \tag{3.23}$$

and

$$C_x(t, \tau) = \Gamma_x(t, \tau) - m_x(t)\, m_x^{\mathrm{T}}(\tau). \tag{3.24}$$

The matrix t function

$$P_x(t) \triangleq C_x(t, t) \tag{3.25}$$

is, for each t, the *covariance matrix* of the random vector x_t [compare with (2.79)]. Now the diagonal elements of $\Gamma_x(t, \tau)$ and $C_x(t, \tau)$ are, respectively, the (auto) correlation and (auto) covariance functions of $(x_t)_i$, $i = 1, \ldots, n$. The off-diagonal elements are, respectively, the (cross) correlation and (cross) covariance functions of $(x_t)_i$ and $(x_t)_j$, $i \neq j$, $i, j = 1, \ldots, n$. If $\{x_t, t \in T\}$ and $\{y_t, t \in T\}$ are two scalar processes,[4] then their **(cross) correlation** and **(cross) covariance** functions are, respectively,

$$\gamma_{xy}(t, \tau) \triangleq \mathscr{E}\{x_t y_\tau\}, \tag{3.26}$$

$$c_{xy}(t, \tau) \triangleq \mathscr{E}\{[x_t - m_x(t)][y_\tau - m_y(\tau)]\}. \tag{3.27}$$

We shall often omit the subscript x on the (auto) correlation and covariance functions (and matrices) when there is no confusion as to which process we are referring.

[4] When speaking of two or more stochastic processes, we always assume that they are defined on the same probability space.

Now the mean value function and correlation function are generally much easier to come by than the probability law of the process. At the same time, they do yield considerable information about the process. Their importance will be emphasized in Sections 4 and 6.

Example 3.7. Consider the process

$$x_t = a + bt \tag{3.8}$$

of Example 3.5. This process has the mean value function

$$m(t) = \mathscr{E}\{a\} + \mathscr{E}\{b\}\, t, \tag{3.28}$$

correlation function

$$\gamma(t, \tau) = \mathscr{E}\{a^2\} + \mathscr{E}\{ab\}(t + \tau) + \mathscr{E}\{b^2\}\, t\tau, \tag{3.29}$$

and covariance function

$$c(t, \tau) = \text{var}\{a\} + \text{cov}\{a, b\}(t + \tau) + \text{var}\{b\}\, t\tau. \tag{3.30}$$

Note that we did not need the joint distribution of a and b, but only their means, variances, and covariance to compute the mean value, correlation, and covariance functions of the x_t process. Suppose a and b are uncorrelated with zero means and unit variances. Then

$$\begin{aligned} m(t) &= 0, \\ \gamma(t, \tau) &= c(t, \tau) = 1 + t\tau. \quad \blacktriangle \end{aligned} \tag{3.31}$$

We now give a brief, formal introduction to the notions of *stationarity* of a stochastic process. Let $\{x_t\,, t \in T\}$ be a stochastic process with a linear parameter set T.[5] The process is said to be **strictly stationary** (or stationary in the **strict sense**) if it has the same probability law as the process $\{x_{t+\tau}\,, t \in T\}$ for any $\tau \in T$. This means that

$$p(x_{t_1}, \ldots, x_{t_n}) = p(x_{t_1+\tau}, \ldots, x_{t_n+\tau}) \tag{3.32}$$

for *all* finite sets $\{t_i\} \in T$ and for any $\tau \in T$. The process is **strictly stationary of order** k if (3.32) holds for $n \leqslant k$ only. Clearly, if (3.32) holds for $n = k$, then it holds for $n < k$ (marginal densities). A special case of (3.32) ($n = 1$) is

$$p(x_t) = p(x_{t+\tau})$$

[5] T is a linear set if $t, \tau \in T$ implies that $t + \tau \in T$. Examples: $T = \{1, 2, \ldots\}$, $T = \{t: t \geqslant 0\}$.

for all τ, which means that the first-order density of the process is independent of t:

$$p(x_t) = p(x, t) = p(x). \tag{3.33}$$

As a result, the mean value function of the process, if it exists, is constant:

$$m_x(t) = \text{constant}. \tag{3.34}$$

It is readily seen that the second-order density [$n = 2$ in (3.32)] can depend only on $(t_1 - t_2)$. This, in turn, implies that the correlation function $\gamma(t_1, t_2)$, if it exists, depends only on $(t_1 - t_2)$, or

$$\gamma_x(t + \tau, t) = \mathscr{E}\{x_{t+\tau}x_t\} = \gamma(\tau). \tag{3.35}$$

Equations (3.34) and (3.35) together imply (and are implied by) the covariance function

$$c_x(t + \tau, t) = c(\tau). \tag{3.36}$$

On the basis of the first- and second-order properties of a strictly stationary process, we define a weaker notion of stationarity. The stochastic process $\{x_t, t \in T\}$ is said to be **weakly stationary** (or stationary in the **wide sense**, or **covariance stationary**) if it has finite second moments[6] and (3.34), (3.35) hold.

Finally, we say that the process $\{x_t, t \in T\}$ has **strictly stationary increments** if the process $\{x_{t+s} - x_t, t \in T\}$ is strictly stationary for every $s \in T$.

Example 3.8. The process

$$x_t = a \cos 2\pi t + b \sin 2\pi t \tag{3.9}$$

of Example 3.5, where a and b are uncorrelated, with zero means and unit variances, is weakly stationary:

$$m(t) = 0,$$
$$\gamma(t + \tau, t) = \cos 2\pi\tau. \quad \blacktriangle$$

3. CONVERGENCE OF RANDOM SEQUENCES

Let $\{x_n, n = 1, 2,...\}$ be a scalar random sequence. There are a number of ways in which the sequence might converge (as $n \to \infty$). We

[6] This means $\mathscr{E}\{|x_t|^2\} < \infty$ for all $t \in T$. This, in turn, implies that the mean value, correlation, and covariance functions exist. See Example 2.9.

define three modes of convergence here. Other modes of convergence can be found in Loeve [9].

The sequence $\{x_n\}$ is said to converge to x **with probability 1** (wp 1) if

$$\lim_{n\to\infty} x_n(\omega) = x(\omega) \tag{3.37}$$

for almost all ω (almost all realizations). That is, (3.37) holds except perhaps on an event A such that $\Pr\{A\} = 0$. We then write

$$\lim x_n = x \qquad \text{wp 1.} \tag{3.38}$$

The sequence $\{x_n\}$ is said to converge to x **in probability** if, for every $\epsilon > 0$,

$$\lim_{n\to\infty} \Pr\{|\, x_n(\omega) - x(\omega)| \geqslant \epsilon\} = 0. \tag{3.39}$$

We write

$$p \lim x_n = x. \tag{3.40}$$

The $\{x_n\}$ sequence is said to converge to x **in mean square** if $\mathscr{E}\{|\, x_n\,|^2\} < \infty$ for all n, $\mathscr{E}\{|\, x\,|^2\} < \infty$, and

$$\lim_{n\to\infty} \mathscr{E}\{|\, x - x_n\,|^2\} = 0. \tag{3.41}$$

We write

$$\text{l.i.m. } x_n = x \tag{3.42}$$

and call x the **limit in the mean** (or **mean square limit**) of $\{x_n\}$.

Now it can be shown that, in general, wp 1 convergence neither implies nor is implied by mean square convergence. This is shown via counterexamples in Bartlett [2, p. 137]. It can be shown that wp 1 convergence implies convergence in probability, Loeve [9, p. 116]. We show here that *mean square convergence* also *implies convergence in probability*. For every $\epsilon > 0$ and rv y with finite second moment, we have

$$\mathscr{E}\{|\, y\,|^2\} = \int_{-\infty}^{\infty} |\, y\,|^2 p(y)\, dy \geqslant \int_{-\infty}^{-\epsilon} |\, y\,|^2 p(y)\, dy + \int_{\epsilon}^{\infty} |\, y\,|^2 p(y)\, dy$$

$$\geqslant \epsilon^2 \left(\int_{-\infty}^{-\epsilon} p(y)\, dy + \int_{\epsilon}^{\infty} p(y)\, dy \right) = \epsilon^2 \Pr\{|\, y\,| \geqslant \epsilon\}.$$

Substituting $x - x_n$ for y, we get

$$\Pr\{|\, x - x_n\,| \geqslant \epsilon\} \leqslant \mathscr{E}\{|\, x - x_n\,|^2\}/\epsilon^2, \tag{3.43}$$

which is called the **Chebychev inequality**. Our assertion follows immediately from this inequality.

A necessary and sufficient condition for mean square convergence is the **Cauchy criterion**:

$$\lim_{n,m\to\infty} \mathscr{E}\{|\, x_n - x_m \,|^2\} = 0. \tag{3.44}$$

That is, the sequence $\{x_n\}$ has a mean square limit if, and only if, (3.44) holds. Necessity is easy to prove and follows from

$$|\, x_n - x_m \,|^2 = |(x_n - x) + (x - x_m)|^2 \leqslant 2\,|\, x_n - x \,|^2 + 2\,|\, x - x_m \,|^2.$$

It can be shown that the Cauchy criterion is also sufficient.[7]

Convergence wp 1 and convergence in probability are more difficult to use than convergence in mean square. On the other hand, the former convergence concepts do not require the existence of second moments, whereas mean square convergence does. In this book, we shall mainly use mean square convergence. We shall indicate, when appropriate, that certain limits also exist wp 1. In that case, it can be shown that both limits must be the same. The properties of mean square (ms) limits are summarized in the following theorem.

Theorem 3.1 (Mean Square Convergence). *$\{x_n\}$, $\{y_n\}$, $\{v_n\}$ are random sequences, z is a random variable, all with finite second moments. $\{c_n\}$ is a sequence of fixed constants, a, b are fixed constants. Let* $x =$ l.i.m. x_n, $y =$ l.i.m. y_n, $v =$ l.i.m. v_n, $c = \lim c_n$.[8] *Then*

(i) l.i.m. $c_n = \lim c_n$;

(ii) l.i.m. $z = z$;

(iii) l.i.m. $zc_n = zc$;

(iv) l.i.m.$(ax_n + by_n) = ax + by$ (*the* "l.i.m." *operation is linear*);

(v) $\mathscr{E}\{x\} = \lim \mathscr{E}\{x_n\}$ ("$\mathscr{E}$" *and* "l.i.m." *operations commute*);

[7] This statement is not completely correct. We assume that all our rv's have density functions, and we define expectation as the Riemann integral (2.27). A Cauchy sequence (sequence satisfying (3.44)) may have as a limit an rv x that does not possess a density, and whose expectation does not exist in the sense of (2.27). If we defined the expectation more generally as a Lebesque integral, then it can be shown (Loeve [9, p. 161]) that the space of rv's with finite second moments, with scalar product $\mathscr{E}\{xy\}$, norm $\mathscr{E}^{1/2}\{|\, x \,|^2\}$, and distance $\mathscr{E}^{1/2}\{|\, x - y \,|^2\}$ is a *complete* metric space (a Hilbert space). In that case, every Cauchy sequence converges and our claim of sufficiency is correct. The pathological case in which the limit of a Cauchy sequence does not have an expectation, as defined in this book, need not concern an engineer.

[8] This is the usual limit, $|\, c - c_n \,| \to 0$ $(n \to \infty)$.

(vi) $\mathscr{E}\{xy\} = \lim \mathscr{E}\{x_n y_n\}$ (*note special case*: $\mathscr{E}\{x^2\} = \lim \mathscr{E}\{x_n^2\}$);

(vii) *if* $\mathscr{E}\{x_n y_n\} = \mathscr{E}\{v_n\}$, *then* $\mathscr{E}\{xy\} = \mathscr{E}\{v\}$.

PROOF: (i), (ii), (iii) are trivially true. (iv) follows from

$$\mathscr{E}\{|\, ax_n + by_n - ax - by\,|^2\} = \mathscr{E}\{|\, a(x_n - x) + b(y_n - y)|^2\}$$
$$\leqslant 2a^2\mathscr{E}\{|\, x_n - x\,|^2\} + 2b^2\mathscr{E}\{|\, y_n - y\,|^2\} \to 0 \qquad (n \to \infty).$$

To prove (v), we use Schwarz's inequality (2.61):

$$|\, \mathscr{E}\{x_n - x\}|^2 \leqslant \mathscr{E}\{|\, x_n - x\,|^2\} \to 0 \qquad (n \to \infty).$$

To prove (vi), we write

$$|\, \mathscr{E}\{xy\} - \mathscr{E}\{x_n y_n\}| = |\, \mathscr{E}\{(y - y_n)\, x\} + \mathscr{E}\{(x - x_n)\, y\} - \mathscr{E}\{(x - x_n)(y - y_n)\}|$$
$$\leqslant |\, \mathscr{E}\{(y - y_n)\, x\}| + |\, \mathscr{E}\{(x - x_n)\, y\}| + |\, \mathscr{E}\{(x - x_n)(y - y_n)\}|,$$

and (vi) follows upon applying Schwarz's inequality to the last three terms. (vii) is true because

$$\mathscr{E}\{v\} = \lim \mathscr{E}\{v_n\} = \lim \mathscr{E}\{x_n y_n\} = \mathscr{E}\{xy\},$$

where we have used (v), hypothesis, and (vi), in that order. ■

Suppose $\{x_n\,,\ n = 1, 2,...\}$ is a random m-vector sequence, where

$$x_n^{\mathrm{T}} = [(x_n)_1\,,...,\,(x_n)_m]. \tag{3.45}$$

We take as the *norm* of a (random) m-vector x the usual Euclidean norm

$$|\, x\,| = (x^{\mathrm{T}}x)^{1/2} = \left(\sum_{i=1}^{m} x_i^2\right)^{1/2}. \tag{3.46}$$

Then the *distance* between the two vectors x and y is

$$|\, x - y\,| = [(x - y)^{\mathrm{T}}\,(x - y)]^{1/2} = \left[\sum_{i=1}^{m} (x_i - y_i)^2\right]^{1/2}. \tag{3.47}$$

We say that the random m-vector sequence $\{x_n\}$ converges in mean square to the random vector x if $\mathscr{E}\{|\, x_n\,|^2\} < \infty$ for all n, $\mathscr{E}\{|\, x\,|^2\} < \infty$, and

$$\lim_{n\to\infty} \mathscr{E}\{|\, x - x_n\,|^2\} = 0. \tag{3.48}$$

It is obvious from the definition of our norm (3.46) that the vector

sequence converges if, and only if, the m sequences $\{(x_n)_i, n = 1,...\}$ of components of x_n converge. We shall use this fact without further comment.

4. MEAN SQUARE CALCULUS

In the study of stochastic differential equations (Chapter 4), we shall need to consider integrals and derivatives of stochastic processes (random functions). In preparation for this, we present here the elements of mean square calculus. This is an extension of the mean square convergence concepts for random sequences (Section 3), to random functions. A detailed mathematical account of this subject can be found in Loeve [9, Chapter X]. We treat here continuity, differentiation, and integration in the mean square sense. These properties and operations can also be defined for the sample functions of the process, that is, with probability one. This more difficult subject is not treated here. We do, however, point out some of the relationships between mean square and wp 1 calculus.

Let $\{x_t(\omega), t \in T\}$ be a scalar, continuous parameter, stochastic process with finite second moments. That is, $\mathscr{E}\{| x_t |^2\} < \infty$ for all $t \in T$. In view of Theorems 2.1 and 2.3, the mean value function $m(t) = \mathscr{E}\{x_t\}$ and the correlation function $\gamma(t, \tau) = \mathscr{E}\{x_t x_\tau\}$ exist for all $t, \tau \in T$. Since the covariance function

$$c(t, \tau) = \gamma(t, \tau) - m(t)\, m(\tau), \tag{3.49}$$

it also exists for all $t, \tau \in T$.

The random function x_t is said to be **continuous in mean square** at $t \in T$ if

$$\text{l.i.m.}_{h \to 0}\, x_{t+h} = x_t \tag{3.50}$$

for $t + h \in T$. Now

$$\mathscr{E}\{| x_{t+h} - x_t |^2\} = \gamma(t + h, t + h) - \gamma(t + h, t) - \gamma(t, t + h) + \gamma(t, t). \tag{3.51}$$

The right-hand side of (3.51) goes to zero with h if $\gamma(t, \tau)$ is continuous at (t, t). Thus, we have a sufficient condition for x_t to be ms continuous at t. This condition is also necessary, since, from Theorem 3.1 (vi), we have

$$\lim_{h, h' \to 0} \mathscr{E}\{x_{t+h} x_{t+h'}\} = \mathscr{E}\{x_t x_t\},$$

which is just

$$\lim_{h,h'\to 0} \gamma(t + h, t + h') = \gamma(t, t). \tag{3.52}$$

We therefore have

Theorem 3.2. *x_t is mean square continuous at $t \in T$ if, and only if, $\gamma(t, \tau)$ is continuous at (t, t).*

Corollary 1. *If $\gamma(t, \tau)$ is continuous at every diagonal point (t, t), then it is continuous at every (t, τ) point.*

PROOF: By Theorem 3.2, x_t is ms continuous at every $t \in T$. Thus, l.i.m. $x_{t+h} = x_t (h \to 0)$, and l.i.m. $x_{\tau+h'} = x_\tau$ $(h' \to 0)$ for every $t, \tau \in T$. In view of Theorem 3.1 (vi), then,

$$\lim_{h,h'\to 0} \mathcal{E}\{x_{t+h}x_{\tau+h'}\} = \mathcal{E}\{x_t x_\tau\}, \tag{3.53}$$

which proves the corollary. ■

Corollary 2. *$\gamma(t, \tau)$ is continuous at (t, t) if, and only if, $m(t)$ is continuous at t and $c(t, \tau)$ is continuous at (t, t).*

PROOF: The *if* part is obvious from (3.49). We now prove the *only if* part. If $\gamma(t, \tau)$ is continuous at (t, t), then, by Theorem 3.2, x_t is ms continuous at t. But then, in view of Theorem 3.1 (v),

$$\lim_{h\to 0} \mathcal{E}\{x_{t+h}\} = \mathcal{E}\{x_t\},$$

which is just

$$\lim_{h\to 0} m(t + h) = m(t). \tag{3.54}$$

Thus, $m(t)$ is continuous at t. This, together with the continuity of $\gamma(t, t)$ and (3.49), proves the continuity of $c(t, \tau)$ at (t, t). ■

We point out that, in view of Theorem 3.1 (iv), the sum of mean square continuous random functions is ms continuous. We note that ms continuity on a t interval does not, in general, imply wp 1 continuity on the interval. That is, a random function may be ms continuous on an interval, and yet some, or even every, sample function can be discontinuous at some points in the interval. Finally, we note that by ms continuity of a random vector function we mean the ms continuity of each of the vector elements.

Let $\{x_n(t, \omega), n = 1, 2,...\}$ be a sequence of random functions. That is,

$\{x_n(t, \omega), t \in T\}$ is a random function for each $n \geqslant 1$. Suppose each member of the sequence is mean square continuous at every $t \in T$, and the sequence converges in mean square, uniformly in t. What can we say about the limit random function?

Theorem 3.3. *The uniform mean square limit of mean square continuous random functions is mean square continuous.*

PROOF (*Indirect*): We have that

$$\underset{n\to\infty}{\text{l.i.m.}}\, x_n(t, \omega) = x(t, \omega) \qquad \text{uniformly in} \quad t \in T.$$

Therefore, by Theorem 3.1 (vi),

$$\lim_{n\to\infty} \mathscr{E}\{x_n(t, \omega)\, x_n(t, \omega)\} = \mathscr{E}\{x(t, \omega)\, x(t, \omega)\}$$

uniformly in t, which is just

$$\lim_{n\to\infty} \gamma_n(t, t) = \gamma(t, t) \tag{3.55}$$

uniformly in t. But since each $x_n(t, \omega)$ is ms continuous at every $t \in T$, then by Theorem 3.2, each $\gamma_n(t, \tau)$ is continuous at every (t, t) point. Therefore, in (3.55) we have the (ordinary) uniform limit of continuous functions, so that $\gamma(t, \tau)$ is continuous at every (t, t) point. But then, by Theorem 3.2, $x(t, \omega)$ is ms continuous at every $t \in T$. ■

(Theorem 3.3 can readily be proved directly, that is, without going to correlation functions and using the theorem in ordinary calculus which is analogous to Theorem 3.3. We leave this proof for the reader.)

The random function x_t is said to be **mean square differentiable** at $t \in T$ if the following limit, which defines the mean square derivative, exists:

$$\underset{h\to 0}{\text{l.i.m.}}(x_{t+h} - x_t)/h = dx_t/dt = \dot{x}_t \tag{3.56}$$

for $t + h \in T$. From (3.56), it is evident that

$$x_{t+h} - x_t = \dot{x}_t h + v,$$

where the remainder v satisfies[9]

$$\mathscr{E}\{v\} = o(h), \qquad \mathscr{E}\{v^2\} = o(h^2).$$

[9] $o(h)$ means that $o(h)/h \to 0$ $(h \to 0)$.

Therefore,

$$\mathcal{E}\{|\, x_{t+h} - x_t \,|^2\} = o(h^2),$$

and we have

Theorem 3.4. *If x_t is mean square differentiable at $t \in T$, then it is mean square continuous at t.*

Theorem 3.5. *x_t is mean square differentiable at $t \in T$, if, and only if, $\partial^2\gamma(t, \tau)/\partial t\, \partial\tau$ exists at (t, t).*

PROOF: In view of the Cauchy criterion (3.44), a sufficient condition for ms differentiability of x_t is the convergence to zero (with h, h') of

$$\mathcal{E}\left\{\left|\frac{x_{t+h} - x_t}{h} - \frac{x_{t+h'} - x_t}{h'}\right|^2\right\}.$$

This does indeed converge to zero if $\partial^2\gamma(t, t)/\partial t\, \partial\tau$ exists, since, in that case,

$$\begin{aligned}\lim_{h,h'\to 0} \mathcal{E}\left\{\frac{x_{t+h} - x_t}{h} \cdot \frac{x_{t+h'} - x_t}{h'}\right\} \\ = \lim_{h,h'\to 0}(hh')^{-1}\{[\gamma(t + h, t + h') - \gamma(t, t + h')] \\ - [\gamma(t + h, t) - \gamma(t, t)]\} = \partial^2\gamma(t, t)/\partial t\, \partial\tau. \qquad (3.57)\end{aligned}$$

This proves the *if* part. To prove the *only if* part, it is sufficient to notice that ms differentiability of x_t implies, via Theorem 3.1 (vi), that

$$\lim_{h,h'\to 0} \mathcal{E}\left\{\frac{x_{t+h} - x_t}{h} \cdot \frac{x_{t+h'} - x_t}{h'}\right\} = \mathcal{E}\{\dot{x}_t \dot{x}_t\}. \qquad (3.58)$$

But this, in view of (3.57), implies the existence of the required derivative of γ. ■

Corollary 1. *If $\partial^2\gamma(t, \tau)/\partial t\, \partial\tau$ exists at every diagonal point (t, t), then it exists at every (t, τ) point.*

PROOF: By Theorem 3.5, x_t is ms differentiable at every $t \in T$. Therefore, in view of Theorem 3.1 (vi),

$$\lim_{h,h'\to 0} \mathcal{E}\left\{\frac{x_{t+h} - x_t}{h} \cdot \frac{x_{\tau+h'} - x_\tau}{h'}\right\} = \mathcal{E}\{\dot{x}_t \dot{x}_\tau\}. \qquad (3.59)$$

But an easy computation, similar to that in (3.57), shows that the foregoing limit is

$$\partial^2\gamma(t, \tau)/\partial t\, \partial\tau. \quad \blacksquare \qquad (3.60)$$

Corollary 2. *$\partial^2\gamma(t, \tau)/\partial t\, \partial\tau$ exists at (t, t) if, and only if, $\dot{m}(t)$ exists at t and $\partial^2 c(t, \tau)/\partial t\, \partial\tau$ exists at (t, t).*

PROOF: The *if* part is obvious from (3.49). To prove the *only if* part, we observe that, by Theorem 3.5, x_t is ms differentiable at t. Then from Theorem 3.1 (v) we have

$$\lim_{h\to 0} \mathscr{E}\left\{\frac{x_{t+h} - x_t}{h}\right\} = \mathscr{E}\{\dot{x}_t\}. \tag{3.61}$$

But this limit is just

$$\dot{m}(t). \tag{3.62}$$

Therefore, $\dot{m}(t)$ exists and since, by hypothesis, $\partial^2\gamma(t, \tau)/\partial t\, \partial\tau$ exists at (t, t), then in view of (3.49)

$$\partial c(t, \tau)/\partial t\, \partial\tau$$

also exists at (t, t). ■

Now suppose x_t is mean square differentiable at every $t \in T$. Then, from Theorem 3.1 (vi),

$$\lim_{h\to 0} \mathscr{E}\left\{\frac{x_{t+h} - x_t}{h} \cdot x_\tau\right\} = \mathscr{E}\{\dot{x}_t x_\tau\}. \tag{3.63}$$

But this limit is nothing more than

$$\partial\gamma(t, \tau)/\partial t. \tag{3.64}$$

Collecting these results together with (3.59)–(3.62), we have

Theorem 3.6. *Let x_t be mean square differentiable at every $t \in T$. Then*

$$m_{\dot{x}}(t) = \mathscr{E}\{\dot{x}_t\} = \frac{d}{dt}\mathscr{E}\{x_t\} = \dot{m}_x(t), \tag{3.65}$$

$$\gamma_{\dot{x}x}(t, \tau) = \mathscr{E}\{\dot{x}_t x_\tau\} = \frac{d}{dt}\mathscr{E}\{x_t x_\tau\} = \frac{\partial\gamma_{xx}(t, \tau)}{\partial t}, \tag{3.66}$$

$$\gamma_{\dot{x}\dot{x}}(t, \tau) = \mathscr{E}\{\dot{x}_t \dot{x}_\tau\} = \frac{d}{dt}\frac{d}{d\tau}\mathscr{E}\{x_t x_\tau\} = \frac{\partial^2\gamma_{xx}(t, \tau)}{\partial t\, \partial\tau}, \tag{3.67}$$

$$c_{\dot{x}x}(t, \tau) = \mathscr{E}\{\dot{x}_t x_\tau\} - \mathscr{E}\{\dot{x}_t\}\,\mathscr{E}\{x_\tau\} = \frac{\partial c_{xx}(t, \tau)}{\partial t}, \tag{3.68}$$

$$c_{\dot{x}\dot{x}}(t, \tau) = \mathscr{E}\{\dot{x}_t \dot{x}_\tau\} - \mathscr{E}\{\dot{x}_t\}\,\mathscr{E}\{\dot{x}_\tau\} = \frac{\partial^2 c_{xx}(t, \tau)}{\partial t\, \partial\tau}. \tag{3.69}$$

That is, the operations of mean square differentiation and expectation commute.

In view of Theorem 3.1 (iv), the sum of ms differentiable functions is ms differentiable, and the ms derivative of the sum equals the sum of the ms derivatives. It should be clear that second- and higher-order ms derivatives can be defined and that these exist if, and only if, appropriate derivatives of the correlation function exist. We add without proof that *ms differentiability implies* wp 1 *continuity. Existence of a second ms derivative implies* wp 1 *existence of the ordinary first derivative*, and so on. Finally, when we say that a random vector function is ms differentiable, we mean that its elements are.

Example 3.9. Let $x_t = at$, where a is a random variable with $\mathscr{E}\{a^4\} < \infty$. Proceeding formally, we have $\dot{x}_t = a$. To verify this, we note that

$$\mathscr{E}\left\{\left|\frac{a(t+h) - at}{h} - a\right|^2\right\} \equiv 0.$$

Now let $y_t = x_t^2$. Formally,

$$\frac{dy_t}{dt} = \frac{dy_t}{dx_t}\frac{dx_t}{dt},$$

which gives $\dot{y}_t = 2x_t a = 2a^2 t$. To verify this,

$$\mathscr{E}\left\{\left|\frac{a^2(t+h)^2 - a^2t^2}{h} - 2a^2t\right|^2\right\} = \mathscr{E}\{a^4\}\, h^2 \to 0 \qquad (h \to 0). \quad \blacktriangle$$

Example 3.10. Let $x_t = \sin at$, where a is the same as in Example 3.9. We show that $\dot{x}_t = a \cos at$:

$$\begin{aligned}
&\mathscr{E}\left\{\left|\frac{\sin a(t+h) - \sin at}{h} - a\cos at\right|^2\right\} \\
&\quad = \mathscr{E}\left\{\left|\frac{\sin at(\cos ah - 1) + \cos at(\sin ah - ah)}{h}\right|^2\right\} \\
&\quad \leqslant 2\mathscr{E}\left\{\frac{\sin^2 at(\cos ah - 1)^2}{h^2}\right\} + 2\mathscr{E}\left\{\frac{\cos^2 at(\sin ah - ah)^2}{h^2}\right\}.
\end{aligned}$$

But, since $\cos\alpha \leqslant 1 + \alpha^2$ and $\sin\alpha \leqslant \alpha + \alpha^2$, the foregoing is

$$\leqslant 4h^2\mathscr{E}\{a^4\} \to 0 \qquad (h \to 0). \quad \blacktriangle$$

Example 3.11. Consider the random function e^{x_t}, where x_t is mean

square continuous and zero-mean Gaussian for every $t \in T$. We show that e^{x_t} is ms continuous. We have the elementary inequality

$$| e^{x_{t+h}} - e^{x_t} | \leqslant \tfrac{1}{2} | x_{t+h} - x_t |[e^{x_{t+h}} + e^{x_t}]. \tag{3.70}$$

Therefore,

$$\begin{aligned}\mathscr{E}\{| e^{x_{t+h}} - e^{x_t} |^2\} &\leqslant \tfrac{1}{4}\mathscr{E}\{| x_{t+h} - x_t |^2 | e^{x_{t+h}} + e^{x_t} |^2\} \\ &\leqslant \tfrac{1}{4}[\mathscr{E}\{| x_{t+h} - x_t |^4\}]^{1/2} [\mathscr{E}\{| e^{x_{t+h}} + e^{x_t} |^4\}]^{1/2}.\end{aligned}$$

But

$$\mathscr{E}\{| x_{t+h} - x_t |^4\} = 3\mathscr{E}\{| x_{t+h} - x_t |^2\}^2$$

by Example 2.6. Also, an easy computation shows that

$$\mathscr{E}\{\exp(vx_t)\} = \exp(v^2\mathscr{E}\{x_t^2\}/2)$$

for every finite real v. (This is just the *moment generating function*. See Parzen [11, p. 11].) Therefore,

$$[\mathscr{E}\{| e^{x_{t+h}} + e^{x_t} |^4\}]^{1/2} \leqslant M < \infty,$$

and so

$$\mathscr{E}\{| e^{x_{t+h}} - e^{x_t} |^2\} \leqslant \frac{3^{1/2}M}{4} \mathscr{E}\{| x_{t+h} - x_t |^2\} \to 0 \qquad (h \to 0). \quad \blacktriangle$$

Let $a, b \in T$, $a \leqslant b$, and partition the interval $[a, b]$:

$$a = t_0 < t_1 < \cdots < t_n = b.$$

Set $\rho = \max_i(t_{i+1} - t_i)$, $t_i \leqslant t_i' < t_{i+1}$. We say that the random function x_t is **mean square Riemann integrable** over $[a, b]$ if the following limit, which then defines the integral, exists:

$$\operatorname*{l.i.m.}_{\rho \to 0} \sum_{i=0}^{n-1} x_{t_i'}(t_{i+1} - t_i) = \int_a^b x_t \, dt. \tag{3.71}$$

Theorem 3.7. *x_t is mean square Riemann integrable over $[a, b]$ if, and only if, $\gamma(t, \tau)$ is Riemann integrable over $[a, b] \times [a, b]$.*

PROOF: Let

$$a = \bar{t}_0 < \bar{t}_1 < \cdots < \bar{t}_m = b$$

be another partition of the interval with

$$\bar{\rho} = \max_j(\bar{t}_{j+1} - \bar{t}_j), \bar{t}_j \leqslant \bar{t}_j' < \bar{t}_{j+1}.$$

According to the Cauchy criterion (3.44), a sufficient condition for the existence of the limit in (3.71) is the convergence to zero (with $\rho, \bar{\rho}$) of

$$\mathscr{E}\left\{\left|\sum_{i=0}^{n-1} x_{t_i'}(t_{i+1} - t_i) - \sum_{j=0}^{m-1} x_{\bar{t}_j'}(\bar{t}_{j+1} - \bar{t}_j)\right|^2\right\}. \tag{3.72}$$

Now

$$\lim_{\rho,\bar{\rho}\to 0} \mathscr{E}\left\{\sum_{i=0}^{n-1} x_{t_i'}(t_{i+1} - t_i) \cdot \sum_{j=0}^{m-1} x_{\bar{t}_j'}(\bar{t}_{j+1} - \bar{t}_j)\right\}$$

$$= \lim_{\rho,\bar{\rho}\to 0} \sum_{i=0}^{n-1} \sum_{j=0}^{m-1} \mathscr{E}\{x_{t_i'} x_{\bar{t}_j'}\}(t_{i+1} - t_i)(\bar{t}_{j+1} - \bar{t}_j) \tag{3.73}$$

is just the Riemann integral of $\gamma(t, \tau)$, if it exists. Thus, if $\gamma(t, \tau)$ is Riemann integrable over $[a, b] \times [a, b]$, the limit in (3.73) exists, and (3.72) converges to zero. This proves the *if* part. To prove the *only if* part, we note that, if x_t is ms Riemann integrable, then the limit in (3.73) exists [Theorem 3.1 (vi)], which means that $\gamma(t, \tau)$ is Riemann integrable. ■

We note that, in fact [Theorem 3.1 (vi)],

$$\int_a^b \int_a^b \gamma(t, \tau)\, dt\, d\tau = \mathscr{E}\left\{\left|\int_a^b x_t\, dt\right|^2\right\},$$

and even more generally

$$\int_a^b \int_c^d \gamma(t, \tau)\, dt\, d\tau = \mathscr{E}\left\{\int_a^b x_t\, dt \cdot \int_c^d x_\tau\, d\tau\right\} \tag{3.74}$$

if x_t is ms Riemann integrable over $[a, b]$ and $[c, d]$. Also, from Theorem 3.1 (v),

$$\int_a^b m(t)\, dt = \mathscr{E}\left\{\int_a^b x_t\, dt\right\}. \tag{3.75}$$

Corollary 1. *$\gamma(t, \tau)$ is Riemann integrable over $[a, b] \times [a, b]$ if, and only if, $m(t)$ is R-integrable over $[a, b]$ and $c(t, \tau)$ is R-integrable over $[a, b] \times [a, b]$.*

PROOF: The *if* part is obvious from (3.49). The *only if* part follows from (3.75) via Theorem 3.7, and from (3.49). ■

From Theorem 3.2, Corollary 1 of that theorem, and from Theorem 3.7, we have

Corollary 2. *x_t is ms Riemann integrable over $[a, b]$ if x_t is ms continuous at every $t \in [a, b]$.*

We now collect computational results (3.74) and (3.75), and their immediate consequences, in

Theorem 3.8. *Let x_t and y_t be mean square Riemann integrable over T, $[a, b] \in T$, $[c, d] \in T$. Then*

$$\mathscr{E}\left\{\int_a^b x_t\,dt\right\} = \int_a^b m(t)\,dt, \tag{3.75}$$

$$\mathscr{E}\left\{\int_a^b x_t\,dt \cdot \int_c^d x_\tau\,d\tau\right\} = \int_a^b \int_c^d \gamma(t, \tau)\,dt\,d\tau, \tag{3.74}$$

$$\operatorname{cov}\left\{\int_a^b x_t\,dt, \int_c^d x_\tau\,d\tau\right\} = \int_a^b \int_c^d c(t, \tau)\,dt\,d\tau. \tag{3.76}$$

That is, the operations of integration and expectation may be interchanged. Also,

$$\left(\int_a^b \int_c^d \gamma_{xy}(t, \tau)\,dt\,d\tau\right)^2 \leqslant \int_a^b \int_a^b \gamma_{xx}(t, \tau)\,dt\,d\tau \cdot \int_c^d \int_c^d \gamma_{yy}(t, \tau)\,dt\,d\tau. \tag{3.77}$$

[To prove (3.77), use (3.74) and Schwarz's inequality.]

We may note that, in view of Theorem 3.1 (iv), the ms Riemann integral is linear and additive in the domain of integration. We state without proof that, *if x_t is ms continuous, then it is also integrable* wp 1. That is, almost all sample functions are integrable. Also, the wp 1 integral equals the ms integral. Finally, by the integral of a random vector function, we mean the vector whose elements are the integrals of the elements of the random vector function.

Suppose x_t is mean square Riemann integrable over $[a, t]$ for every $t \in [a, b]$. Then

$$y_t = \int_a^t x_\tau\,d\tau \tag{3.78}$$

is a random function (of t) defined on $[a, b]$. Clearly [see (3.74)], $\mathscr{E}\{y_t^2\} < \infty$ for every $t \in [a, b]$. We now study some of the properties of this random function.

Theorem 3.9. *Let x_t be ms Riemann integrable over $[a, t]$ for every $t \in [a, b]$. Then y_t (3.78) is ms continuous on $[a, b]$. Furthermore, if x_t is ms continuous on $[a, b]$, then y_t is ms differentiable on (a, b) and $\dot{y}_t = x_t$.*

PROOF (*continuity*):

$$\mathscr{E}\left\{\left|\int_a^{t+h} x_\tau\,d\tau - \int_a^t x_\tau\,d\tau\right|^2\right\} = \mathscr{E}\left\{\left|\int_t^{t+h} x_\tau\,d\tau\right|^2\right\}$$

$$= \int_t^{t+h}\int_t^{t+h} \gamma(s,\tau)\,ds\,d\tau \to 0 \qquad (h \to 0),$$

since the last Riemann integral is a continuous function of its upper limit.

PROOF (*differentiability*):

$$\mathscr{E}\left\{\left|h^{-1}\int_t^{t+h} x_\tau\,d\tau - x_t\right|^2\right\} = (h^2)^{-1}\,\mathscr{E}\left\{\left|\int_t^{t+h}(x_\tau - x_t)\,d\tau\right|^2\right\}$$

$$= (h^2)^{-1}\int_t^{t+h}\int_t^{t+h} \mathscr{E}\{(x_s - x_t)(x_\tau - x_t)\}\,ds\,d\tau$$

$$\leqslant (h^2)^{-1}\left(\int_t^{t+h} \mathscr{E}^{1/2}\{(x_\tau - x_t)^2\}\,d\tau\right)^2.$$

Since x_t is ms continuous on $[a, b]$, by Theorem 3.2 $\mathscr{E}^{1/2}\{(x_\tau - x_t)^2\}$ is continuous on $[a, b]$. Then by the mean value theorem for Riemann integrals,

$$\int_t^{t+h} \mathscr{E}^{1/2}\{(x_\tau - x_t)^2\}\,d\tau = h\mathscr{E}^{1/2}\{(x_{\tau_0} - x_t)^2\}, \qquad \tau_0 \in [t, t+h].$$

Therefore,

$$\mathscr{E}\left\{\left|h^{-1}\int_t^{t+h} x_\tau\,d\tau - x_t\right|^2\right\} \leqslant \mathscr{E}\{(x_{\tau_0} - x_t)^2\} \to 0 \qquad (h \to 0). \quad \blacksquare$$

Theorem 3.10 (Fundamental Theorem of ms Calculus). *Let $\dot{x}_t$ be ms Riemann integrable on $[a, b]$. Then*

$$x_t - x_a = \int_a^t \dot{x}_\tau\,d\tau$$

with probability 1.

PROOF: We show that

$$\mathscr{E}\left\{\left|x_t - x_a - \int_a^t \dot{x}_\tau\,d\tau\right|^2\right\} = 0,$$

and then the conclusion follows via Example 2.5. Now

$$\mathscr{E}\left\{\left| x_t - x_a - \int_a^t \dot{x}_\tau \, d\tau \right|^2\right\} = \mathscr{E}\{(x_t - x_a)^2\} - 2\int_a^t \mathscr{E}\{(x_t - x_a)\,\dot{x}_\tau\}\, d\tau + \int_a^t \int_a^t \mathscr{E}\{\dot{x}_s \dot{x}_\tau\}\, ds\, d\tau.$$

By Theorem 3.6, this equals

$$\gamma_{xx}(t, t) - 2\gamma_{xx}(t, a) + \gamma_{xx}(a, a) - 2\int_a^t \left[\frac{\partial \gamma_{xx}(t, \tau)}{\partial \tau} - \frac{\partial \gamma_{xx}(a, \tau)}{\partial \tau}\right] d\tau + \int_a^t \int_a^t \frac{\partial^2 \gamma_{xx}(s, \tau)}{\partial s\, \partial \tau}\, ds\, d\tau.$$

Carrying out the foregoing integrations shows that this is zero. ■

Example 3.12. Let $z_t = 2a^2 t$, where a is an rv with $\mathscr{E}\{a^4\} < \infty$. Formally,

$$\int_0^t z_\tau \, d\tau = a^2 t^2.$$

We verify this:

$$\int_0^t z_\tau \, d\tau = \text{l.i.m. } 2a^2 \sum_{i=0}^{n-1} \tau_i^2 (\tau_{i+1} - \tau_i)$$

$$= 2a^2 \lim \sum_{i=0}^{n-1} \tau_i^2 (\tau_{i+1} - \tau_i),$$

the last equality following from Theorem 3.1 (iii). But the last limit is the limit of an ordinary Riemann sum and is just $t^2/2$. Thus the formal result is verified. Compare this with Example 3.9. ▲

5. INDEPENDENCE, CONDITIONING, THE BROWNIAN MOTION PROCESS

Two stochastic processes $\{x_t, t \in T\}$ and $\{y_t, t \in T\}$ are said to be **independent** if, for all finite parameter sets $\{t_i\} \in T$, the random vectors

$$[x_{t_1}, \ldots, x_{t_n}]^T, \qquad [y_{t_1}, \ldots, y_{t_n}]^T$$

are independent.

We may consider the conditional probability density

$$p(x_{t_1}, \ldots, x_{t_n} \mid y_{t_1}, \ldots, y_{t_n}) \tag{3.79}$$

or

$$p(x_{t_n} \mid y_{t_1}, \ldots, y_{t_n}) \tag{3.80}$$

and associated expectations. These have already been defined in Section 2.5, and their properties are given in Theorems 2.8 and 2.9. If $\{y_t, t \in T\}$ is a continuous parameter process, we can consider

$$p(x_t \mid y_\tau, \tau \in T), \tag{3.81}$$

the probability density of x_t, given the realization $y_\tau, \tau \in T$ of the $\{y_t, t \in T\}$ process. The rigorous definition of the conditional density in (3.81) is beyond the scope of this book. Let $T = [a, b]$. Partition this interval

$$a = t_1 < t_2 < \cdots < t_n = b$$

and let $\rho = \max_i(t_{i+1} - t_i)$. Then we shall consider the conditional density in (3.81) as the formal limit

$$p(x_t \mid y_\tau, \tau \in T) = \operatorname*{l.i.m.}_{\substack{\rho \to 0 \\ n \to \infty}} p(x_t \mid y_{t_1}, \ldots, y_{t_n}), \tag{3.82}$$

assuming the limit exists. Clearly, if the $\{x_t\}$ and $\{y_t\}$ processes are independent, then the densities in (3.79), (3.80), and (3.81) are equal to

$$p(x_{t_1}, \ldots, x_{t_n}), \qquad p(x_{t_n}), \qquad p(x_t),$$

respectively.

A continuous parameter process $\{x_t, t \in T\}$ has **independent increments** if, for all finite sets $\{t_i : t_i < t_{i+1}\} \in T$, the random variables (vectors)

$$x_{t_2} - x_{t_1}, x_{t_3} - x_{t_2}, \ldots, x_{t_n} - x_{t_{n-1}}$$

are independent. It follows directly from Example 3.6 that the joint characteristic function

$$\varphi_{x_{t_1}, \ldots, x_{t_n}}(u_1, \ldots, u_n) = \varphi_{x_{t_1}}(u_1 + \cdots + u_n) \prod_{i=2}^{n} \varphi_{x_{t_i} - x_{t_{i-1}}}(u_i + \cdots + u_n). \tag{3.83}$$

Therefore, to specify the probability law of a process with independent increments, it suffices to specify the distribution of x_t and $x_t - x_\tau$

for all $t > \tau \in T$. The $\{x_t\}$ process is said to have **stationary independent increments** (see Section 2) if, in addition,

$$x_{t+h} - x_{\tau+h}$$

has the same distribution as $x_t - x_\tau$ for every $t > \tau \in T$ and every $h > 0$.

A stochastic process that is of great importance in theory and applications, and that will play a prominent role in this book, is the **Brownian motion** (or **Wiener**, or **Wiener–Lévy**) process. A continuous parameter process $\{x_t, t \geqslant 0\}$ is a Brownian motion process if

(i) $\{x_t, t \geqslant 0\}$ has stationary independent increments;

(ii) for every $t \geqslant 0$, x_t is normally distributed;

(iii) for every $t \geqslant 0$, $\mathscr{E}\{x_t\} = 0$;

(iv) $\Pr\{x_0 = 0\} = 1$.

Examples of Brownian motion processes include particles undergoing Brownian motion and thermal noise in electric circuits.

It follows from Theorem 2.11 that $x_t - x_\tau$ is also normally distributed for every $t, \tau \geqslant 0$. In view of (3.83), to specify the probability law of the Brownian motion process, it remains to specify the distribution of $x_t - x_\tau$ for all $t > \tau \geqslant 0$. Since $x_t - x_\tau$ is Gaussian, its distribution is determined by its mean and variance. Clearly,

$$\mathscr{E}\{x_t - x_\tau\} = 0. \tag{3.84}$$

It can be shown (Parzen [11, p. 28]), from the fact that the Brownian motion process has stationary independent increments, that

$$\operatorname{var}\{(x_t - x_\tau)\} = \sigma^2(t - \tau), \qquad t > \tau, \tag{3.85}$$

where σ^2 (called the *variance parameter*) is an empirical positive constant.

Let us compute the correlation function of the Brownian motion process. (Its mean value function is identically zero.) Let $t > \tau$:

$$\begin{aligned}\gamma(t, \tau) &= \mathscr{E}\{x_t x_\tau\} = \mathscr{E}\{[(x_t - x_\tau) + x_\tau]\, x_\tau\} \\ &= \mathscr{E}\{(x_t - x_\tau)\, x_\tau\} + \mathscr{E}\{x_\tau^2\} = 0 + \sigma^2\tau.\end{aligned}$$

[In view of (iv), $x_\tau = x_\tau - x_0$.] Therefore,

$$\gamma(t, \tau) = \sigma^2 \min(t, \tau) \qquad \text{for all} \quad t, \tau > 0.^{10} \tag{3.86}$$

[10] $\min(t, \tau)$ equals t if $t \leqslant \tau$, and equals τ if $\tau \leqslant t$.

Note that $\min(t, \tau)$ is continuous at every (t, τ), but $\partial^2 \min(t, \tau)/\partial t\, \partial \tau$ exists at no (t, t). Therefore, in view of Theorem 3.2, the Brownian motion process is mean square continuous on $[0, \infty)$. In view of Theorem 3.7, it is ms Riemann integrable on every t interval. However, according to Theorem 3.5, it is ms differentiable nowhere. It can also be shown (Doob [5, p. 393]) that almost all sample functions of the Brownian motion process are continuous (continuous wp 1). Otherwise, the sample functions of the Brownian motion process are very irregular. They are also differentiable nowhere wp 1. Furthermore, they are of unbounded variation wp 1.

Example 3.13 (*Brownian Motion as Limit of a Random Walk*). Consider the random walk of Example 3.1 with discrete random steps as defined in (3.1) and (3.2). We shall derive a continuous state space, continuous parameter process by considering infinitesimal steps Δx in infinitesimal time increments Δt, and passing to the $\Delta x = \Delta t = 0$ limit. We first compute the characteristic function of a random step w_i. By expanding e^{iuw_i} and averaging, it is easy to compute

$$\varphi_{w_i}(u) = pe^{iu\Delta x} + qe^{-iu\Delta x}.$$

In view of (3.3), and using the fact that the w_i are independent,

$$\varphi_{x_n}(u) = [pe^{iu\Delta x} + qe^{-iu\Delta x}]^n.$$

Now in time t there will be $n = t/\Delta t$ steps; therefore,

$$\varphi_{x_t}(u) = [pe^{iu\Delta x} + qe^{-iu\Delta x}]^{t/\Delta t}. \tag{3.87}$$

Let $p = q = \frac{1}{2}$. By differentiating the characteristic function [see (2.37)], we compute the mean and variance of x_t

$$\mathscr{E}\{x_t\} = 0, \qquad \operatorname{var}\{x_t\} = t(\Delta x)^2/\Delta t.$$

Now we wish to pass to the $\Delta x = \Delta t = 0$ limit. In order to get a sensible result, we must require that $(\Delta x)^2/\Delta t$ have a limit, say σ^2. In that way the limiting process will have finite variance. With this restriction, then, (3.87) becomes

$$\varphi_{x_t}(u) = \{\tfrac{1}{2}[\exp(iu\sigma\, \Delta t^{1/2}) + \exp(-iu\sigma\, \Delta t^{1/2})]\}^{t/\Delta t}. \tag{3.88}$$

We note that for Δt small, $\Delta x = O(\Delta t^{1/2})$, so that the displacement Δx

must be orders of magnitude larger than Δt, the time interval in which it occurs. Taking the logarithm in (3.88), we find that

$$\ln \varphi_{x_t}(u) = -\tfrac{1}{2}u^2\sigma^2 t + O(\Delta t),$$

and in the limit

$$\varphi_{x_t}(u) = \exp(-\tfrac{1}{2}u^2\sigma^2 t). \tag{3.89}$$

Comparing this with (2.44) of Example 2.6, we see that in the limit x_t is Gaussian with zero mean and variance $\sigma^2 t$. Since the random walk has stationary independent increments (w_i), our limiting process x_t does also. Our limiting process is, therefore, a Brownian motion process. ▲

6. GAUSSIAN PROCESSES

A stochastic process $\{x_t, t \in T\}$ is a **Gaussian** (or **normal**) process if the probability law [(3.5), (3.6), or (3.7)] of the process is normal. Since the Gaussian density function is completely characterized by the first two moments, it is clear that one can obtain the probability law of a normal process from its mean value and covariance functions. These, in turn, can be obtained from the first- and second-order densities of the process. For the same reason, a weakly stationary Gaussian process is strictly stationary.

Example 3.14. Consider the process

$$x_t = a + bt$$

of Example 3.5, where the rv's a and b are Gaussian, uncorrelated, with zero means and unit variances. We saw in Example 3.5 that the probability law of the x_t process can be obtained via Theorem 2.11; the x_t process is Gaussian. From (3.10), we have

$$\mathscr{E}[x_{t_1}, \ldots, x_{t_n}]^{\mathrm{T}} = 0,$$

$$P \equiv \mathscr{E}\begin{bmatrix} x_{t_1} \\ \vdots \\ x_{t_n} \end{bmatrix}[x_{t_1}, \ldots, x_{t_n}]$$

$$= \begin{bmatrix} 1 + t_1^2, & 1 + t_1 t_2, & \ldots, & 1 + t_1 t_n \\ 1 + t_2 t_1, & 1 + t_2^2, & \ldots, & 1 + t_2 t_n \\ \vdots & \vdots & \vdots & \vdots \\ 1 + t_n t_1, & 1 + t_n t_2, & \ldots, & 1 + t_n^2 \end{bmatrix}.$$

Now the mean value and covariance functions of this process have been computed in Example 3.7 (3.31) and are

$$m(t) = 0, \qquad c(t, \tau) = \gamma(t, \tau) = 1 + t\tau.$$

It is easy to see that the covariance matrix P can be generated from the covariance function $c(t, \tau)$. The probability law of the x_t process can now be written in terms of the characteristic function

$$\varphi_{x_{t_1},\ldots,x_{t_n}}^{(u)} = \exp(-\tfrac{1}{2}u^{\mathrm{T}}Pu),$$

where $u^{\mathrm{T}} = [u_1, \ldots, u_n]$. In a similar way, it follows by Theorem 2.11 that the process in (3.9) of Example 3.5 is Gaussian if a and b are jointly Gaussian. The correlation function of that process, in case a and b are uncorrelated, is given in Example 3.8. As was noted in Example 3.8, that process is weakly stationary. Let the reader convince himself that it is also strictly stationary. ▲

Example 3.15. The Brownian motion process is a normal process. This can be seen from the following reasoning. Given n times $t_1 < t_2 < \cdots < t_n$ one can write

$$x_{t_k} = (x_{t_k} - x_{t_{k-1}}) + (x_{t_{k-1}} - x_{t_{k-2}}) + \cdots + x_{t_1}$$

for any $1 \leqslant k \leqslant n$. Now the Brownian motion increments on the right-hand side are independent and Gaussian, and are therefore jointly Gaussian. As a result, the x_{t_k}'s are linear combinations of jointly normal rv's, and by Theorem 2.11, are themselves jointly normal. ▲

We have already seen (here and in Chapter 2) that Gaussian random variables and, as a result, Gaussian processes have many nice mathematical properties. One of these outstanding properties is embodied in Theorem 2.11, which essentially says that when we perform linear algebraic operations on Gaussian processes we get back a Gaussian process. This has already been demonstrated in Example 3.14. A very important application of this theorem is to linear random difference equations. The difference equation

$$x_{n+1} = x_n + w_n\,; \qquad n = 0, 1, \ldots; \qquad x_0 = 0, \tag{3.90}$$

which we have already encountered in Example 3.1, generates a Gaussian

process (random sequence) if the w_i are independent Gaussian random variables. This is easy to see, since the solution of (3.90) is

$$x_n = \sum_{i=0}^{n-1} w_i ,$$

and an argument identical to that in Example 3.15 shows that the $\{x_n , n = 0, 1,...\}$ process is Gaussian. We shall have much more to say about random difference equations in Section 9 of this chapter and throughout this book.

Another outstanding property of Gaussian processes is contained in the following theorem.

Theorem 3.11. *Let* $\{x_n , n = 1, 2,...\}$ *be a Gaussian random sequence, and let* l.i.m. $x_n = x$. *Then* x *is Gaussian.*

PROOF: To prove that x is Gaussian, we show that its characteristic function is given by

$$\varphi_x(u) = \exp(iu\mathscr{E}\{x\} - \tfrac{1}{2}u^2 \operatorname{var}\{x\}).$$

Since x_n is Gaussian for each n (Theorem 2.10), we have

$$\varphi_{x_n}(u) = \exp(iu\mathscr{E}\{x_n\} - \tfrac{1}{2}u^2 \operatorname{var}\{x_n\}).$$

Thus we have to show that (*a*) $\mathscr{E}\{x_n\} \to \mathscr{E}\{x\}$, (b) $\operatorname{var}\{x_n\} \to \operatorname{var}\{x\}$, and (c) $\varphi_{x_n} \to \varphi_x$ $(n \to \infty)$. But (*a*) is just the statement of Theorem 3.1 (v); (b) follows from (a) and Theorem 3.1 (vi). As for (c),

$$\begin{aligned} |\varphi_{x_n} - \varphi_x| &= |\mathscr{E}\{\exp(iux_n) - \exp(iux)\}| \\ &\leqslant \mathscr{E}\{|\exp(iux_n) - \exp(iux)|\} \\ &= \sqrt{2}\,\mathscr{E}\{|1 - \cos u(x_n - x)|^{1/2}\} \leqslant |u|\,\mathscr{E}\{|x_n - x|\} \to 0. \quad \blacksquare \end{aligned}$$

The consequence of this theorem is that ms derivatives and integrals (Section 4) of Gaussian processes are themselves Gaussian processes. This will have important application in the study of linear stochastic (random) differential equations in Chapter 4, and in our study of filtering theory.

The engineering importance of Gaussian processes stems from the fact that many physical processes are approximately Gaussian. We have already studied the Brownian motion process which, among other things, provides a good model for particles undergoing Brownian motion and for stellar motion in the universe. It often turns out that when a large number of

small independent random effects are superimposed, then, regardless of their individual distribution, the distribution of the sum of these effects is approximately Gaussian. This is essentially the content of the central limit theorem (see, for example, Papoulis [10, p. 266]). Let x_i be independent random variables. Consider the sum

$$x = x_1 + \cdots + x_n .$$

Then under certain general conditions, and if x is properly scaled, the distribution of x approaches the Gaussian distribution as $n \to \infty$. We have already performed such a central limiting operation when we obtained the Brownian motion process as the limit of a random walk in Example 3.13. It is actually amazing how well the Gaussian density is approximated for very modest n in some cases. Papoulis [10, p. 268] has pictured an example in which the x_i are uniformly and identically distributed. The approximation is surprisingly good for $n = 3$.

Because of the content of the central limit theorem just discussed, a Gaussian process is often a good model for the "noise" in physical devices such as radio transmitters and receivers, radars, and control systems. This will be formalized in Section 8.

7. MARKOV PROCESSES

The reader is familiar with the prominent role played by ordinary differential equations in the analysis of deterministic dynamical systems. The importance of ordinary differential equations stems in large part from certain nice mathematical properties that they enjoy. The differential equation

$$dx(t)/dt = f(x(t))$$

says that the rate of change of x at time t depends only on x at t (*now*), and *not* on $x(\tau)$, $\tau < t$. As a result of this, and as is well-known, for $t_1 < t_2$,

$$x(t_2) = g(t_2 \,;\, x(t_1), t_1).$$

That is, the solution at t_2 is a function of $x(t_1)$ and does not depend on $x(\tau)$, $\tau < t_1$.

This property of ordinary differential equations has its stochastic analog in a class of processes called Markov processes. A discrete or continuous parameter stochastic process $\{x_t , t \in T\}$ is called a **Markov process** if, for any finite parameter set $\{t_i : t_i < t_{i+1}\} \in T$, and for every real λ,

$$\Pr\{x_{t_n}(\omega) \leqslant \lambda \mid x_{t_1}, \ldots, x_{t_{n-1}}\} = \Pr\{x_{t_n}(\omega) \leqslant \lambda \mid x_{t_{n-1}}\}. \tag{3.91}$$

The Markov property (3.91) says that the probability law of the process in the future, once it is in a given state, does not depend on how the process arrived at the given state. This property is sometimes referred to as the generalized causality principle: the future can be predicted from a knowledge of the present. As we shall subsequently see, the Markov property is a basic assumption that is made in the study of stochastic dynamical systems.

For a continuous parameter process, Eq. (3.91) implies that for $t_1 < t_2$ and all λ,

$$\Pr\{x_{t_2}(\omega) \leqslant \lambda \mid x_\tau\,,\ \tau \leqslant t_1\} = \Pr\{x_{t_2}(\omega) \leqslant \lambda \mid x_{t_1}\}. \tag{3.92}$$

The Markov property (3.91) can be written in terms of density functions

$$p(x_{t_n} \mid x_{t_1}\,,\ldots,\, x_{t_{n-1}}) = p(x_{t_n} \mid x_{t_{n-1}}), \tag{3.93}$$

where $t_1 < t_2 < \cdots < t_n$. Equation (3.92) becomes

$$p(x_{t_2} \mid x_\tau\,,\ \tau \leqslant t_1) = p(x_{t_2} \mid x_{t_1}). \tag{3.94}$$

Now consider the joint density function $p(x_{t_n}\,,\ x_{t_{n-1}}\,,\ldots,\, x_{t_1})$. By definition of the conditional density function (2.100), this can be written as

$$p(x_{t_n}\,,\ldots,\, x_{t_1}) = p(x_{t_n} \mid x_{t_{n-1}}\,,\ldots,\, x_{t_1})\, p(x_{t_{n-1}}\,,\ldots,\, x_{t_1}).$$

Now, using the Markov property (3.93),

$$p(x_{t_n}\,,\ldots,\, x_{t_1}) = p(x_{t_n} \mid x_{t_{n-1}})\, p(x_{t_{n-1}}\,,\ldots,\, x_{t_1}).$$

Continuing this process, we finally obtain

$$p(x_{t_n}\,,\ldots,\, x_{t_1}) = p(x_{t_n} \mid x_{t_{n-1}}) \cdot p(x_{t_{n-1}} \mid x_{t_{n-2}}) \cdots p(x_{t_2} \mid x_{t_1}) \cdot p(x_{t_1}). \tag{3.95}$$

Therefore, the probability law of a Markov process can be specified by specifying $p(x_t)$ and $p(x_t \mid x_\tau)$ for all $t > \tau \in T$. The conditional densities $p(x_t \mid x_\tau)$ are called the **transition probability densities** of the Markov process.

Example 3.16. The random sequence $\{x_n\,,\ n = 1, 2,\ldots\}$, where the x_i are mutually independent, is a Markov sequence since

$$p(x_n \mid x_{n-1}\,,\ldots,\, x_1) = p(x_n \mid x_{n-1}) = p(x_n). \quad \blacktriangle$$

Example 3.17. The Brownian motion process is a Markov process, since

$$\begin{aligned} p(x_{t_n} \mid x_{t_{n-1}}, \ldots, x_{t_1}) &= p(x_{t_n} - x_{t_{n-1}} + x_{t_{n-1}} - x_0 \mid x_{t_{n-1}} - x_0, \ldots, x_{t_1} - x_0) \\ &= p(x_{t_n} - x_{t_{n-1}} + x_{t_{n-1}} - x_0 \mid x_{t_{n-1}} - x_0) \\ &= p(x_{t_n} \mid x_{t_{n-1}}). \end{aligned}$$

Since the Brownian motion process is also Gaussian (Example 3.15), it is a *Gauss–Markov* process. Clearly, the process of Brownian motion increments is a Markov process (by Example 3.16), since the increments are independent. ▲

Example 3.18. The process of Example 3.14 is *not* a Markov process. We can write

$$x_{t_n} = x_{t_{n-1}} + b(t_n - t_{n-1})$$

and

$$x_{t_n} = x_{t_{n-1}} + (x_{t_{n-1}} - x_{t_{n-2}})(t_n - t_{n-1})/(t_{n-1} - t_{n-2}).$$

Therefore, we see that $p(x_{t_n} \mid x_{t_{n-1}})$ depends on the density of b, whereas $p(x_{t_n} \mid x_{t_{n-1}}, x_{t_{n-2}})$ is a Dirac delta function. We can, however, construct a Markov process by increasing the dimension of the process. Let the reader show that the vector process

$$x_t = a + bt, \qquad y_t = a + b(t - 1)$$

is Markov. ▲

Example 3.19. Consider the difference equation

$$x_{n+1} = x_n + w_n\,; \qquad n = 0, 1, \ldots, \tag{3.90}$$

where the w_i are independent Gaussian random variables, and the initial condition x_0 is Gaussian, independent of w_i. The sequence generated by (3.90) is clearly Markov since, *given* x_n, x_{n+1} depends only on w_n, which is independent of $x_{n-1}, \ldots, x_0$. The sequence $\{x_n, n = 0, 1, \ldots\}$ is therefore Gauss–Markov. We compute its transition probability densities via Theorem 2.7. Given x_n, x_{n+1} is a linear function of w_n and, from Theorem 2.7,

$$p(x_{n+1} \mid x_n) = p_{w_n}(x_{n+1} - x_n). \tag{3.96}$$

The density on the right-hand side is the density of w_n evaluated at $x_{n+1} - x_n$. ▲

In Example 3.19, we wrote down the transition probability densities between adjacent time steps for the sequence generated by difference equation (3.90). It turns out that all other transition probability densities can be obtained from these via an integral difference equation. We develop that equation now. We have

$$p(x_n \mid x_{n-2}) = \int p(x_n , x_{n-1} \mid x_{n-2})\, dx_{n-1} \tag{3.97}$$

as a marginal density [see (2.53)]. Now the integrand can be written

$$\begin{aligned} p(x_n , x_{n-1} \mid x_{n-2}) &= p(x_n \mid x_{n-1} , x_{n-2})\, p(x_{n-1} \mid x_{n-2}) \\ &= p(x_n \mid x_{n-1})\, p(x_{n-1} \mid x_{n-2}), \end{aligned}$$

where the last equality follows from the Markov property. Therefore,

$$p(x_n \mid x_{n-2}) = \int p(x_n \mid x_{n-1})\, p(x_{n-1} \mid x_{n-2})\, dx_{n-1} . \tag{3.98}$$

Equation (3.98) is called the **Chapman–Kolmogorov** equation. Note that the x_n sequence may be a vector sequence. In that case, the integral in (3.98) is over E^n. Note also that (3.98) holds for a continuous parameter process as well. In that case $n = t_n$, and so on.

It is clear that we can replace x_{n-2} in (3.98) by x_m , $m \leqslant n - 2$, and

$$p(x_n \mid x_m) = \int p(x_n \mid x_{n-1})\, p(x_{n-1} \mid x_m)\, dx_{n-1} . \tag{3.99}$$

The transition probability density $p(x_n \mid x_{n-1})$ is the kernel of (3.99). The initial condition for (3.99) is $p(x_m \mid x_m) = \delta(x_m)$, a Dirac delta function.

Now taking the expectation in (3.99) (see Theorem 2.8), we have

$$p(x_n) = \int p(x_n \mid x_{n-1})\, p(x_{n-1})\, dx_{n-1} . \tag{3.100}$$

Thus the probability density of x_n , given only the density of x_0 (the initial condition), also satisfies the Chapman–Kolmogorov equation. The density $p(x_n)$ is of great interest in the study of discrete stochastic dynamical systems, such as the one of Example 3.19. We shall make use of (3.100) in the study of stochastic difference equations in Section 9.

All the stochastic processes we shall encounter in the remainder of this book are Markov. As we have already seen, Markov processes arise in the study of stochastic difference equations (discrete dynamical systems).

They also arise in the study of stochastic differential equations (continuous dynamical systems). There is a large body of theory associated with Markov processes and voluminous applications to various physical problems. These topics are not germane to the subject of this book, and we do not treat them here. Formal treatments of Markov processes with many applications may be found in Bharucha–Reid [3], Parzen [11], Papoulis [10], and Cox and Miller [4]. Mathematical treatments may be found in Doob [5], Itô [7], and Dynkin [6].

8. WHITE NOISE

In modeling any physical process or device, be it as simple as a pendulum or as sophisticated as a spacecraft guidance and control system, the engineer proceeds by delineating the variables describing the process and then connecting them via certain causal relationships or physical laws. He then simulates his model to determine how well it can predict the physical process. He often finds that his predictions are inexact. At this point he may, if he is willing, remodel the process in order to obtain better predictability. He may or may not succeed. If he repeats this process many times, he will reach a point where further modeling does not improve the prediction. This may be because there are actually unpredictable fluctuations in the process for which no causal relationships exist, or because he has reached the level of random errors in the instruments he is using to measure the physical process, or both. In any case, he has reached the "noise level." At this point, if our engineer is ambitious, he might attempt to determine the statistics of the noise (its probability law) by repeated experimentation with the process or device and statistical testing.

A useful concept, and in many cases a good model for the "noise" we discussed previously, is white noise. A **white random sequence** $\{x_n, n = 1, 2,...\}$ is a Markov sequence for which

$$p(x_k \mid x_l) = p(x_k) \qquad (k > l). \tag{3.101}$$

That is, all the x_k's are mutually independent. As a result, knowing the realization of x_l in no way helps in predicting what x_k will be. A white sequence is completely random or totally unpredictable. If the x_k's are all normally distributed, the $\{x_n\}$ sequence is called a **white Gaussian random sequence**. We have already noted (Section 6) that noise due to the superposition of a large number of small, independent, random effects is often Gaussian, and that this is the statement of the central limit theorem. The importance and usefulness of white Gaussian

sequences stem from this fact. The motivation for calling the sequence white will become apparent when we discuss the analogous continuous parameter process.

Let $\{x_n, n = 1, 2,...\}$ be a white Gaussian random vector sequence. Because the sequence is Gaussian, its probability law is specified by the mean value vector

$$\mathscr{E}\{x_n\}, \qquad \text{all} \quad n \geqslant 1, \tag{3.102}$$

and the covariance matrix

$$\mathscr{E}\{(x_n - \mathscr{E}\{x_n\})(x_m - \mathscr{E}\{x_m\})^{\mathrm{T}}\}, \qquad \text{all} \quad n, m \geqslant 1. \tag{3.103}$$

Because the sequence is white,

$$\mathscr{E}\{(x_n - \mathscr{E}\{x_n\})(x_m - \mathscr{E}\{x_m\})^{\mathrm{T}}\} = Q_n \delta_{nm}, \tag{3.104}$$

where

$$\delta_{nm} = \begin{cases} 1, & n = m \\ 0, & n \neq m \end{cases} \tag{3.105}$$

is the Kronecker delta and Q_n is a positive semidefinite matrix. Therefore, the probability law of the sequence is specified by the mean value vector and the covariance matrix Q_n, all $n \geqslant 1$.

We wish to extend the notion of white noise to the continuous parameter case. Our motivation is as follows. Since the white Gaussian sequence serves as a good model for the noise in discrete physical processes or dynamical systems, its continuous analog may serve as a good model for the noise in continuous dynamical systems, that is, those which evolve in continuous time. We might formally define a white process $\{x_t, t \in T\}$ as a Markov process for which

$$p(x_t \mid x_\tau) = p(x_t), \qquad t > \tau \in T. \tag{3.106}$$

That is, the x_t's are mutually independent for all $t \in T$. If the x_t's are normally distributed for each $t \in T$, then the process is a white Gaussian process.

This formal definition, though expressing the properties we wish the white Gaussian process to possess, is not useful in characterizing the process. We attempt to characterize the process indirectly. Consider a scalar zero-mean stationary Gaussian process with correlation function

$$\gamma^{\rho}(t + \tau, t) = \sigma^2(\rho/2)\, e^{-\rho|\tau|}. \tag{3.107}$$

Now, for ρ large, this process approximates the properties we wish for

our white Gaussian process. We note that if ρ is an integer, then the sequence $\{\gamma^{\rho_i}/\sigma^2, \rho_1 < \rho_2 < \cdots\}$ defines the Dirac delta function. Therefore,

$$\gamma^{\infty} = \sigma^2\delta(\tau). \tag{3.108}$$

We therefore *formally* define a **white Gaussian process** $\{x_t, t \in T\}$ as a Gaussian process with

$$\mathcal{E}\{(x_t - \mathcal{E}\{x_t\})(x_\tau - \mathcal{E}\{x_\tau\})^{\mathrm{T}}\} = Q(t)\,\delta(t - \tau), \tag{3.109}$$

where $Q(t)$ is a positive semidefinite covariance matrix, and $\delta(t - \tau)$ is the Dirac delta function. For obvious reasons this process is often called a *delta-correlated* process. Since the Dirac delta function is not an ordinary function, we see that, although useful, the white Gaussian process is a mathematical fiction.

To gain more insight into the properties of white Gaussian noise, consider the *power spectral density function*,[11] which is defined as the Fourier transform of the correlation function

$$f(\omega) = \int_{-\infty}^{\infty} e^{-i\tau\omega}\gamma(t + \tau, t)\,d\tau.^{12} \tag{3.110}$$

For the process with correlation function (3.107), the power spectral density function is computed to be

$$f^{\rho}(\omega) = \frac{\sigma^2}{1 + (\omega/\rho)^2}. \tag{3.111}$$

With $\rho = \infty, f^{\rho}(\omega) = \sigma^2$, a positive constant. This is the origin of the adjective *white* in white Gaussian process. It is used by analogy with *white light*, which contains all frequency components. But a constant power spectral density requires infinite power and, therefore, white Gaussian noise is not physically realizable. If the power spectral density is essentially flat up to very high frequencies, the process is called a *wide band* process. Its correlation function is almost a delta function. The white Gaussian process is then a useful approximation.

Example 3.20 (White Gaussian process as limit of a white Gaussian sequence).[13] Let $\{x_t^{(\Delta)}, t \geqslant 0\}$ be the Gaussian vector process defined by

$$x_t^{(\Delta)} = x_n, \qquad n\Delta \leqslant t < (n + 1)\Delta, \tag{3.112}$$

[11] The spectral density and related concepts useful in the frequency domain analysis of stationary stochastic processes are treated in Papoulis [10, Chapter 10] and Parzen [11, Chapter 3].

[12] Here ω is the frequency and is not to be confused with the probability parameter.

[13] This example is due to Kalman *et al.* [8].

where $\{x_n, n = 0, 1,...\}$ is a zero-mean, white Gaussian random vector sequence with

$$\mathscr{E}\{x_k x_l^{\mathrm{T}}\} = Q\delta_{kl},$$

where Q is a constant matrix, and $1/\Delta$ is an integer. A sample function of this process is shown in Fig. 3.3. Consider the random vector defined

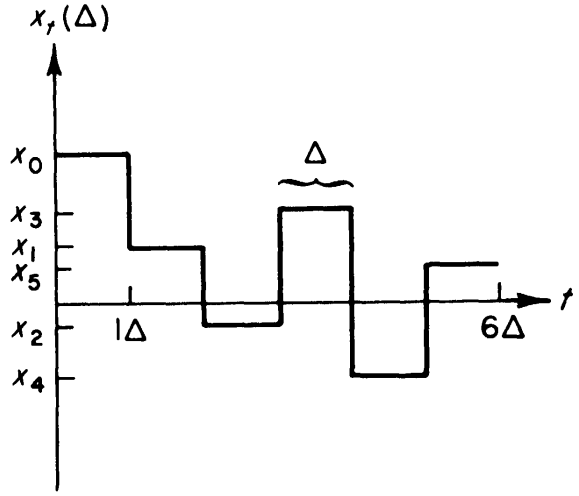


FIG. 3.3. White noise sequence.

for each sample function of the $x_t^{(\Delta)}$ process (wp 1) by the ordinary (Riemann) integral

$$y = \int_0^1 x_t^{(\Delta)}\, dt.$$

Clearly, $\mathscr{E}\{y\} = 0$. Now compute

$$\begin{aligned}\operatorname{var}\{|y|\} &= \mathscr{E}\left\{\left[\int_0^1 x_t^{(\Delta)}\, dt\right]^{\mathrm{T}}\left[\int_0^1 x_t^{(\Delta)}\, dt\right]\right\} \\ &= \operatorname{tr}\int_0^1\int_0^1 \mathscr{E}\{x_t^{(\Delta)}x_\tau^{(\Delta)\mathrm{T}}\}\, dt\, d\tau \\ &= \operatorname{tr}\sum_{i=1}^{1/\Delta} Q\,\Delta^2 = \operatorname{tr} Q\,\Delta.^{14}\end{aligned} \tag{3.113}$$

This expression goes to zero with Δ if Q is kept constant. This says that the effect of white noise in the differential equation

$$dy/dt = x_t^{(\Delta)}$$

reduces to zero with Δ. Now this is physically absurd. Hence, to keep $\operatorname{var}\{|y|\}$ constant as $\Delta \to 0$, we must replace Q by Q/Δ. This means

[14] tr A is the trace of the matrix A.

that the amplitude of the random steps in the sample functions of the $x_t^{(\Delta)}$ process increases as $\Delta^{-1/2}$, whereas the area under a random step tends to zero as $\Delta^{1/2}$. This example gives a heuristic justification for (3.109). Q/Δ goes over to $Q \cdot$ (Dirac delta). White Gaussian noise sample functions may be formally regarded as delta functions of vanishingly small area. ▲

Example 3.21 (*White Gaussian noise as derivative of Brownian motion*). Although we know (Section 5) that the Brownian motion process is not differentiable in any sense, let us proceed formally. Let $\{x_t, t \geqslant 0\}$ be a Brownian motion process. We know that its covariance function is (3.86)

$$c_x(t, \tau) = \sigma^2 \min(t, \tau).$$

From Theorem 3.6, the covariance function of the process $\{dx_t/dt, t \geqslant 0\}$ is

$$c_{\dot{x}}(t, \tau) = \frac{\partial^2 c_x(t, \tau)}{\partial t \, \partial \tau}.$$

Therefore,

$$c_{\dot{x}}(t, \tau) = \sigma^2 \frac{\partial^2 \min(t, \tau)}{\partial t \, \partial \tau}.$$

Now

$$\min(t, \tau) = \begin{cases} \tau, & \tau < t \\ t, & \tau > t \end{cases}$$

so that

$$\frac{\partial}{\partial t} \min(t, \tau) = \begin{cases} 0, & \tau < t \\ 1, & \tau > t \end{cases}$$

which is the Heaviside unit step function (in τ), and its derivative with respect to τ is the Dirac delta $\delta(t - \tau)$. As a result,

$$c_{\dot{x}}(t, \tau) = \sigma^2 \delta(t - \tau),$$

and $\dot{x}_t$ is a delta-correlated process. Thus, formally, white Gaussian noise is the derivative of Brownian motion. ▲

9. STOCHASTIC DIFFERENCE EQUATIONS

Many physical processes and devices subject to random disturbances, whose state can be represented as a finite-dimensional vector, can be modeled via a vector difference equation. Let x_k be the n-dimensional

state vector at time t_k and w_k the m-dimensional vector of random disturbances ($m \leqslant n$). We can write such a difference equation as

$$x_{k+1} = f(x_k, w_{k+1}, t_k), \qquad k = 0, 1,..., \tag{3.114}$$

where f is a real, in general nonlinear, n-vector function, which, we suppose, is a continuously differentiable function of its arguments. Equation (3.114) is a nonlinear *stochastic vector difference equation* and represents (is sometimes called) a *discrete stochastic dynamical system.* The disturbance w_k is often called the *random noise input* to the system. The sequence $\{w_k, k = 1, 2,...\}$ is a random vector sequence, and, as a result, so is $\{x_k, k = 0, 1,...\}$. The initial condition for (3.114), that is, x_0, may be a given constant vector or a random vector with a specified distribution. The probability law of $\{w_k, k = 1, 2,...\}$ must also be specified. We have already encountered a special case of (3.114) in Example 3.19.

If the random input w_k is absent from (3.114), we have an ordinary difference equation, and we speak of x_k as its solution. In the presence of the random input, it is the density of x_k, $p(x_k)$, which is usually of interest. From it we can compute $\mathscr{E}\{x_k\}$, $\text{var}\{x_k\}$, and so on, which give us much information about the statistical behavior of x_k. Probabilities such as $\Pr\{|x_k| \leqslant M\}$ are also often of interest and can be computed from $p(x_k)$.

If the probability law of $\{w_k\}$ is arbitrary, little can be said about the dynamical system (3.114). Let $\{w_k\}$ be a white Gaussian sequence, independent of the initial condition x_0. Then, given x_k, x_{k+1} depends only on w_{k+1}, which is independent of x_{k-1},..., x_0. Therefore, the solution of (3.114) is a *Markov* sequence.[15]

Suppose that $n = m$ and (3.114) can be solved for w_{k+1}. That is, for fixed x_k and t_k, $f(x_k, \cdot, t_k)$ has an inverse, say f^{-1}, which is continuously differentiable. Then, according to Theorem 2.7, we have the transition probability density

$$p(x_{k+1} \mid x_k) = p_{w_{k+1}}(f^{-1}(x_k, x_{k+1}, t_k)) \left\| \frac{\partial f^{-1}}{\partial x_{k+1}} \right\|. \tag{3.115}$$

With these transition probability densities in hand, $p(x_k)$, $k = 1, 2,...$ can be computed from (3.100), namely,

$$p(x_{k+1}) = \int p(x_{k+1} \mid x_k)\, p(x_k)\, dx_k\,. \tag{3.116}$$

$p(x_0)$ is given. If $x_0 = c$, a fixed constant, then $p(x_0) = \delta(x_0 - c)$.

[15] Actually, $\{x_k, k = 0, 1,...\}$ is Markov if $\{w_k, k = 1, 2,...\}$ is just white, not necessarily Gaussian.

Now if $w_{k+1} \sim N(0, Q_{k+1})$ and $Q_{k+1} > 0$, then

$$p_{w_{k+1}}(f^{-1}(x_k, x_{k+1}, t_k)) = [(2\pi)^{n/2} \mid Q_{k+1} \mid^{1/2}]^{-1}$$

$$\cdot \exp\{-\tfrac{1}{2}(f^{-1}(x_k, x_{k+1}, t_k))^{\mathrm{T}} Q_{k+1}^{-1}(f^{-1}(x_k, x_{k+1}, t_k))\}. \quad (3.117)$$

If Q_{k+1} is only semidefinite, say of rank r, then according to Anderson [1, Theorem 6, p. 344] there exists a nonsingular $n \times n$ matrix A_{k+1} such that

$$A_{k+1} Q_{k+1} A_{k+1}^{\mathrm{T}} = \begin{bmatrix} I_{r\times r}, & 0 \\ 0, & 0 \end{bmatrix}. \quad (3.118)$$

Let $v_{k+1} = A_{k+1} w_{k+1}$. If we partition

$$\begin{aligned} v_{k+1}^{\mathrm{T}} &= [\overset{(1\times r)}{v_{k+1}^{(1)\mathrm{T}}}, \quad v_{k+1}^{(2)\mathrm{T}}], \\ A_{k+1}^{-1} &= [\overset{(n\times r)}{B_{k+1}}, \quad C_{k+1}], \end{aligned} \quad (3.119)$$

then, in view of (3.118), $v_{k+1}^{(1)} \sim N(0, I_{r\times r})$ and

$$\mathscr{E}\{v_{k+1}^{(2)} v_{k+1}^{(2)\mathrm{T}}\} = 0. \quad (3.120)$$

But then, according to Example 2.5, $v_{k+1}^{(2)} = 0$ wp 1, and therefore

$$w_{k+1} = B_{k+1} v_{k+1}^{(1)}. \quad (3.121)$$

If we substitute (3.121) in (3.114), then we have a problem with noise of lower dimension than the state, where the noise covariance matrix is positive definite. This is a special case of the following analysis.

Suppose $m < n$ (which is an important case), the matrix $\partial f/\partial w_{k+1}$ is of rank m, and $w_{k+1} \sim N(0, Q_{k+1})$, $Q_{k+1} > 0$.[16] Then, with the aid of an implicit function theorem (and perhaps a change of variables), Eq. (3.114) can be put in the form

$$\begin{aligned} x_{k+1}^{(1)} &= f^{(1)}(x_k, w_{k+1}, t_k), \\ x_{k+1}^{(2)} &= f^{(2)}(x_k, x_{k+1}^{(1)}, t_k), \end{aligned} \quad (3.122)$$

where x_k is partitioned,

$$x_k^{\mathrm{T}} = [\overset{(1\times m)}{x_k^{(1)\mathrm{T}}}, \quad x_k^{(2)\mathrm{T}}],$$

[16] If ever rank $\partial f/\partial w_{k+1} <$ rank Q_{k+1}, then we can make them equal by augmenting the state x_k. Add dummy state variables $(x_{k+1})_i = (w_{k+1})_i$.

and $\partial f^{(1)}/\partial w_{k+1}$ is nonsingular. Now we have

$$p(x_{k+1} \mid x_k) = p(x_{k+1}^{(1)}, x_{k+1}^{(2)} \mid x_k)$$
$$= p(x_{k+1}^{(2)} \mid x_{k+1}^{(1)}, x_k)\, p(x_{k+1}^{(1)} \mid x_k). \tag{3.123}$$

Since $f^{(1)}$ has an inverse, we can apply Theorem 2.7 to the first of equations (3.122) and write $p(x_{k+1}^{(1)} \mid x_k)$ as in (3.115) and (3.117). Now

$$p(x_{k+1}^{(2)} \mid x_{k+1}^{(1)}, x_k) = \delta(x_{k+1}^{(2)} - f^{(2)}), \tag{3.124}$$

a Dirac delta function. The specification of $p(x_{k+1} \mid x_k)$ in (3.123) is complete.

Example 3.22. Consider the linear system

$$x_{k+1} = \Phi_{k+1,k}x_k + \bar{\Gamma}_{k+1}\bar{w}_{k+1}, \tag{3.125}$$

where x_k is the n-vector state, $\Phi_{k+1,k}$ is the $n \times n$ (nonsingular) *state transition matrix*, $\bar{\Gamma}_{k+1}$ is $n \times m$, and $\bar{w}_{k+1}$ is $m \times 1$. Suppose $\bar{\Gamma}_{k+1}\bar{w}_{k+1}$ has a singular covariance matrix

$$\bar{\Gamma}_{k+1}\bar{Q}_{k+1}\bar{\Gamma}_{k+1}^{\mathrm{T}}.$$

Then, as we have already noted below (3.117),

$$\bar{\Gamma}_{k+1}\bar{w}_{k+1} = \Gamma_{k+1}w_{k+1},$$

where w_{k+1} is of dimension $r \leqslant m$ and has a positive definite covariance matrix. Γ_{k+1} is $n \times r$ and of rank r. System (3.125) therefore becomes

$$x_{k+1} = \Phi_{k+1,k}x_k + \Gamma_{k+1}w_{k+1}. \tag{3.126}$$

Now partition

$$\Gamma_{k+1} = \begin{bmatrix} \Gamma_{k+1}^{(1)} \\ \Gamma_{k+1}^{(2)} \end{bmatrix},$$

where $\Gamma_{k+1}^{(1)}$ is $r \times r$, and assume, without loss of generality, that $\Gamma^{(1)}$ is nonsingular. Let $y_k = T_k x_k$, where T_k is the nonsingular transformation

$$T_k = \begin{bmatrix} I^{(r\times r)}, & 0 \\ -\Gamma_k^{(2)}\Gamma_k^{(1)-1}, & I \end{bmatrix}, \qquad T_k^{-1} = \begin{bmatrix} I, & 0 \\ \Gamma_k^{(2)}\Gamma_k^{(1)-1}, & I \end{bmatrix}.$$

Let

$$\Psi_{k+1,k} = T_{k+1}\Phi_{k+1,k}T_k^{-1}.$$

Then system (3.126) transforms into

$$\begin{bmatrix} y_{k+1}^{(1)} \\ y_{k+1}^{(2)} \end{bmatrix} = \Psi_{k+1,k} \begin{bmatrix} y_k^{(1)} \\ y_k^{(2)} \end{bmatrix} + \begin{bmatrix} \Gamma_{k+1}^{(1)} w_{k+1} \\ 0 \end{bmatrix}, \tag{3.127}$$

which is of the form (3.122). ▲

An important special case of (3.114) is the *linear* discrete stochastic dynamical system

$$x_{k+1} = \Phi_{k+1,k} x_k + \Gamma_{k+1} w_{k+1}, \qquad k = 0, 1, \ldots . \tag{3.128}$$

x_k is the n-vector state of the system, $\Phi_{k+1,k}$ is the $n \times n$ state transition matrix, Γ_{k+1} is the $n \times m$ noise coefficient matrix, and $\{w_k, k = 1, 2, \ldots\}$ is a white, Gaussian, random m-vector sequence, $w_k \sim N(0, Q_k)$. The initial condition x_0 is assumed independent of the $\{w_k\}$ sequence.

With the machinery we have just developed, we can compute

$$p(x_{k+1} \mid x_k)$$

and obtain $p(x_k)$, $k = 1, 2, \ldots$ from (3.116). From (3.116), we can compute difference equations for

$$\hat{x}_k = \mathscr{E}\{x_k\} \tag{3.129}$$

and

$$P_k = \mathscr{E}\{(x_k - \hat{x}_k)(x_k - \hat{x}_k)^{\mathrm{T}}\}. \tag{3.130}$$

This is a laborious algebraic exercise, especially if we consider the possibility that $\Gamma_{k+1} w_{k+1}$ may have a singular covariance matrix and have to use the transformation developed in Example 3.22. We leave that exercise for the reader. Instead we notice that, if x_0 is a fixed constant, or if x_0 is Gaussian, $x_0 \sim N(\hat{x}_0, P_0)$, then, by Theorem 2.11, Eq. (3.128) generates a *Gauss–Markov* sequence. Then $x_k \sim N(\hat{x}_k, P_k)$; $\hat{x}_k$ and P_k completely determine $p(x_k)$.

We compute difference equations for $\hat{x}_k$ and P_k directly from (3.128). Taking the expectation in (3.128), we have

$$\hat{x}_{k+1} = \Phi_{k+1,k} \hat{x}_k, \tag{3.131}$$

since w_k is zero-mean. Subtracting (3.131) from (3.128),

$$x_{k+1} - \hat{x}_{k+1} = \Phi_{k+1,k}(x_k - \hat{x}_k) + \Gamma_{k+1} w_{k+1}.$$

Squaring this and taking the expectation,

$$P_{k+1} = \Phi_{k+1,k} P_k \Phi_{k+1,k}^{\mathrm{T}} + \Gamma_{k+1} Q_{k+1} \Gamma_{k+1}^{\mathrm{T}} ,$$

since x_k and w_{k+1} are independent. To summarize, the mean vector and covariance matrix of x_k evolve according to

$$\begin{aligned} \hat{x}_{k+1} &= \Phi_{k+1,k} \hat{x}_k , \\ P_{k+1} &= \Phi_{k+1,k} P_k \Phi_{k+1,k}^{\mathrm{T}} + \Gamma_{k+1} Q_{k+1} \Gamma_{k+1}^{\mathrm{T}} . \end{aligned} \tag{3.132}$$

Note that Eqs. (3.132) describe the evolution of $\hat{x}_k$ and P_k even if x_0 is not Gaussian (or constant). In that case, however, the $\{x_k\}$ sequence, though Markov, is not Gaussian.

We observe that the effect of the noise in (3.128) is to add uncertainty to the system. $\Gamma_{k+1} Q_{k+1} \Gamma_{k+1}^{\mathrm{T}}$ is positive semidefinite and tends to make the eigenvalues of P_{k+1} larger. This, in turn, tends to "flatten" the density $p(x_{k+1})$. If the noise is absent from (3.128), we can think of w_{k+1} as a random vector with a density that is a Dirac delta function concentrated at 0. But this is the limiting case of $N(0, Q_{k+1})$ as $Q_{k+1} \to 0$. Therefore, we need only set $Q_{k+1} \equiv 0$ in (3.132). We could have, of course, rederived this result from (3.128) after setting $w_{k+1} \equiv 0$. We also could have applied Theorem 2.7 and obtained $p(x_{k+1})$, and subsequently its parameters [(3.132) with $Q_{k+1} \equiv 0$], from (3.128). Alternatively, we could compute $p(x_{k+1} \mid x_k)$ (it is a delta function) from (3.128) and use (3.116). The reader may pursue the details of these alternate derivations.

Now if, in addition to the absence of noise in (3.128), the initial condition x_0 is certain (say $x_0 = c$), then $p(x_k) = \delta(x_k - c)$, a delta function for all k. In that case, Eqs. (3.132) reduce to

$$\hat{x}_{k+1} = \Phi_{k+1,k} \hat{x}_k , \qquad \hat{x}_0 = c, \qquad P_k \equiv 0, \tag{3.133}$$

a completely deterministic (nonrandom) system.

Consider the scalar, nonlinear, stochastic difference equation with *additive* noise

$$x_{k+1} = f(x_k) + g(x_k)\, w_{k+1} , \qquad k = 0, 1, \ldots, \tag{3.134}$$

where $\{w_k\}$ is a white Gaussian sequence, $w_k \sim N(0, q_k)$, $p(x_0)$ is given, and x_0 is assumed independent of $\{w_k\}$. $\{x_k\}$ is clearly a Markov sequence. Assuming $g(x_k) \neq 0$, we can compute $p(x_k)$ using the theory that we have developed previously. Instead, however, we attempt to determine the equations of evolution of $\hat{x}_k$ (3.129) and P_k (3.130) directly from

(3.134). We note that these two parameters *do not* determine $p(x_k)$, since x_k is not, in general, Gaussian. They do, however, give information about the mean path of system (3.134) and the dispersion about that path.

Taking the expectation in (3.134) and assuming that it, and subsequent expectations, exist,

$$\hat{x}_{k+1} = \mathscr{E}\{f(x_k)\}. \tag{3.135}$$

Subtracting (3.135) from (3.134), squaring, and averaging,

$$P_{k+1} = \mathscr{E}\{f^2(x_k)\} - \mathscr{E}^2\{f(x_k)\} + \mathscr{E}\{g^2(x_k)\}\, q_{k+1}\,. \tag{3.136}$$

Now the expectations on the right-hand sides of (3.135) and (3.136) require the knowledge of $p(x_k)$ for their evaluation. We can evaluate these expectations approximately by making certain assumptions about $p(x_k)$. Suppose $p(x_k)$ is assumed symmetric and "close to the mean." Then its third central moment is zero, and fourth and higher-order central moments might be neglected. If f and g are sufficiently smooth, we can expand f, f^2, and g^2 in Taylor series around $\hat{x}_k$ and then average. For example,

$$f(x_k) = f(\hat{x}_k) + f_x(\hat{x}_k)(x_k - \hat{x}_k) + \tfrac{1}{2} f_{xx}(\hat{x}_k)(x_k - \hat{x}_k)^2 + \cdots$$

and

$$\mathscr{E}\{f(x_k)\} \cong f(\hat{x}_k) + \tfrac{1}{2} f_{xx}(\hat{x}_k)\, P_k\,,$$

since $\mathscr{E}\{x_k - \hat{x}_k\} = 0$, and in view of our assumption about $p(x_k)$.[17] We thus obtain approximate difference equations for $\hat{x}_k$ and P_k :

$$\begin{aligned} \hat{x}_{k+1} &= f(\hat{x}_k) + (P_k/2) f_{xx}(\hat{x}_k), \\ P_{k+1} &= P_k f_x^{\,2}(\hat{x}_k) - \tfrac{1}{4} P_k^{\,2} f_{xx}^2(\hat{x}_k) \\ &\quad + q_{k+1}[g^2(\hat{x}_k) + P_k g_x^{\,2}(\hat{x}_k) + P_k g(\hat{x}_k)\, g_{xx}(\hat{x}_k)]. \end{aligned} \tag{3.137}$$

Note that these difference equations are nonlinear and *coupled*, unlike (3.132) for the linear system. They do reduce to (3.132) if f is linear and g is independent of x.

We point out that other approximating assumptions on $p(x_k)$ are possible. We could suppose that $p(x_k)$ is close to the mean and Gaussian

[17] Strictly speaking, we should use the Taylor series with a remainder and assume, for example, that f_{xxx} is uniformly bounded, or bounded in mean square. Then we would need only that the sixth central moment of $p(x_k)$ be small, and we would not need to say anything about very high-order moments. Recall from Example 2.6 that, even if $p(x_k)$ is close to the mean (variance small), the moments can grow without bound as the order of the moment gets large.

and retain fourth central moments, writing these in terms of the second central moments as in Example 2.6. We note that the analysis can be readily extended to the vector case. The notation, in that case, becomes rather cumbersome.

REFERENCES

1. T. W. Anderson, "Introduction to Multivariate Statistical Analysis." Wiley, New York, 1958.
2. M. S. Bartlett, "An Introduction to Stochastic Processes." Cambridge Univ. Press, London and New York, 1956.
3. A. T. Bharucha-Reid, "Elements of the Theory of Markov Processes and their Applications." McGraw-Hill, New York, 1960.
4. D. R. Cox and H. D. Miller, "The Theory of Stochastic Processes." Wiley, New York, 1965.
5. J. L. Doob, "Stochastic Processes." Wiley, New York, 1953.
6. E. B. Dynkin, "Markov Processes." Vols. 1 and 2. Academic Press, New York, 1965.
7. K. Itô, Lectures on Stochastic Processes (mimeographed notes), Tata Institute of Fundamental Research, Bombay, India, 1961.
8. R. E. Kalman, T. S. Englar, and R. S. Bucy, Fundamental Study of Adaptive Control Systems, ASD-TR-61-27, Vol. 1, 1962.
9. M. Loeve, "Probability Theory." Van Nostrand, Princeton, New Jersey, 1963.
10. A. Papoulis, "Probability, Random Variables, and Stochastic Processes." McGraw-Hill, New York, 1965.
11. E. Parzen, "Stochastic Processes." Holden-Day, San Francisco, California, 1962.